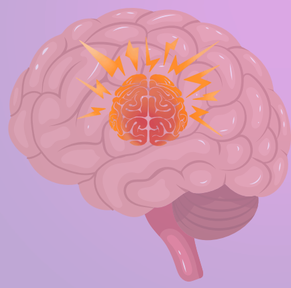
ALZHEIMER'S DETECTION USING 3D CNN–VIT WITH EXPLAINABLE AI



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MOTIVATION

- Alzheimer’s disease is a progressive neurodegenerative disorder where early detection can support timely treatment planning and patient care.
- Structural MRI can reveal important brain changes associated with Alzheimer’s disease, such as hippocampal atrophy and ventricular enlargement.
- Manual MRI assessment can be time-consuming and observer-dependent, while deep learning can provide supportive evidence to assist radiologists in making more confident clinical decisions.
- 3D analysis uses the entire MRI volume, preserving more information than 2D slices.

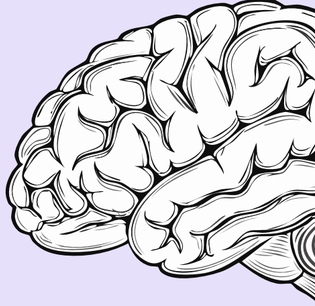
Related Works

Paper	Model Type Used	Dataset Used	Limitations	Accuracy/ Performance
Ayush Pandey, Ayushma Pandey et al. (2025) [1]	Pretrained ViT	Kaggle Alzheimer MRI (5760 images)	Strong Dataset imbalance; no GradCAM explainability; not a true 3D volumetric MRI approach	Train 82%, Val 72%
Yagis et al. (2021) [2]	2D CNNs — VGG16 (frozen + fine-tuned)	ADNI (100 AD + 100 CN), OASIS, PPMI	Slice-level train/test split leaks slices of the same subject into both sets	Slice-level: 95.2–96.9% subject-level: 66.5–70.1%
Atefe Aghaei and Mohsen Ebrahimi Moghaddam. (2025) [3]	Ensemble Transfer Learning + 3D Models + ViT-GAN + 3D-CNN	ADNI T1-weighted MRI scans + demographic data	Limited progressive CN samples, high computational cost due to 3D ViT-GAN.	Accuracy: 0.85 F1-score: 0.86

Related Works

Paper	Model Type Used	Dataset Used	Limitations	Accuracy/ Performance
Wen et al. (2020) [4]	2D-slice, 3D-patch, ROI & 3D-subject CNNs	ADNI + AIBL (T1-weighted MRI)	Survey found >50% of reviewed papers had data leakage (slice-level / augmentation);results not reproducible	Subject-level AD vs CN accuracy 85-88%, leaky pipelines overestimate accuracy.
Lyes Saad Saoud and Hasan AlMarzouqi. (2024)[5]	138 × 3D ViTs + Deep Belief Network Ensemble	MALPEM-ADNI (532 MRI)	Small dataset, slice level split, only 60% accuracy for 3-class	AD vs CN 90%, sMCI vs pMCI AUROC 95%
K Velu and N Jaisankar. (2025)[6]	EfficientNet-B0 + Swin Transformer + DCGAN	OASIS (augmented to 2600/class)	2D slice-based; no external dataset testing; synthetic data bias due to DCGAN	95.34% accuracy

RESEARCH GAP



1. Many existing studies use slice-level data splitting, potentially inflating reported performance through patient leakage. [2,4,5]

2. Lack of leakage-free subject-level benchmarking.

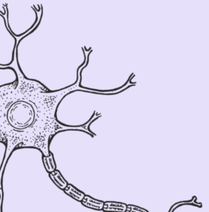
3. Limited use of 3D Explainable AI (XAI) to validate clinically relevant brain regions.

✗ **Slice Level Split**

Same patient in train and test, giving inflated results

✓ **Subject-level Split**

No patient straddles the split, honest accuracy



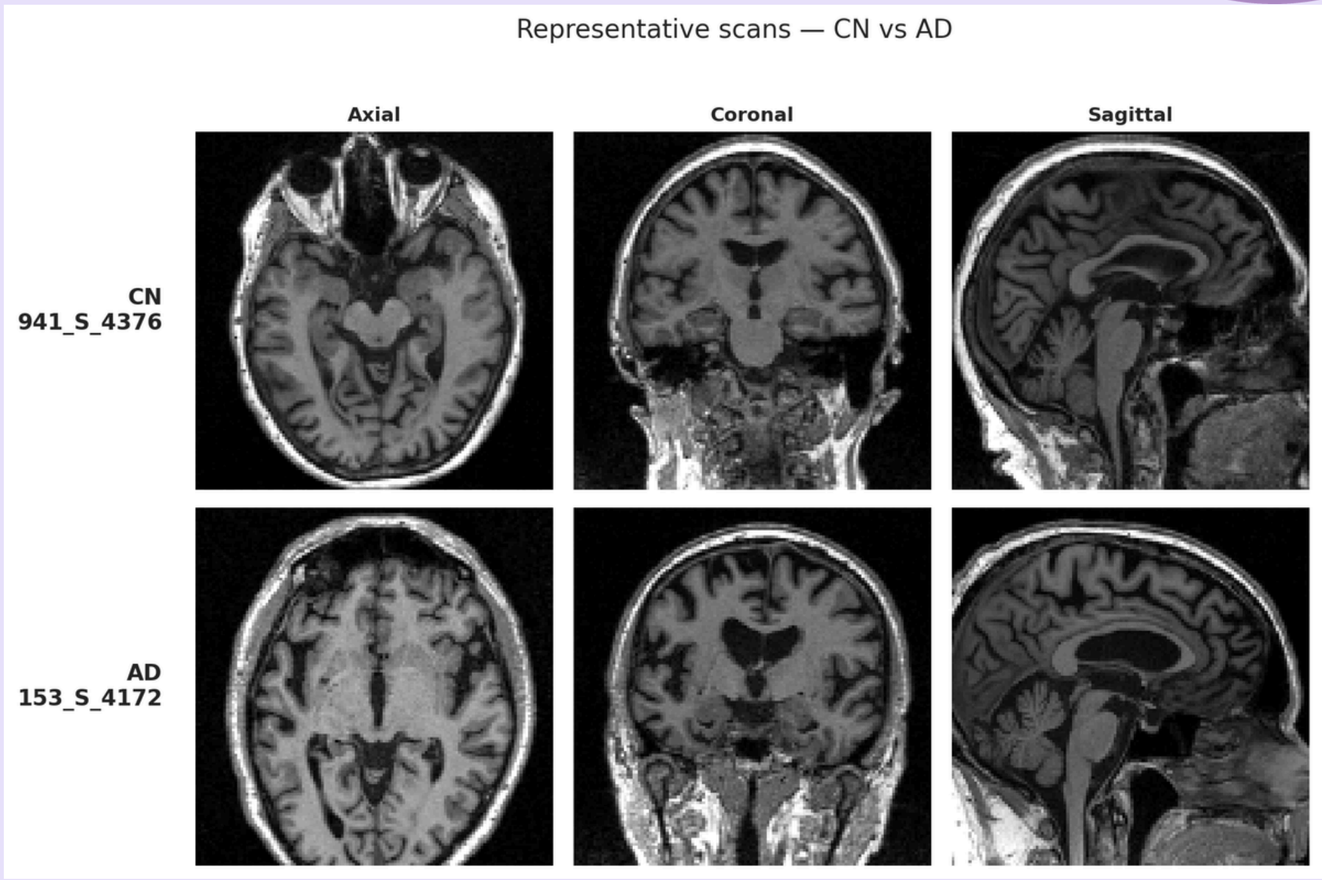
RESEARCH QUESTIONS

- RQ1 — Hybrid Performance
  Can a 3D CNN–Transformer hybrid model outperform single CNN or Transformer models?

- RQ2 — Explainability
  Can 3D XAI methods highlight AD-relevant brain regions?

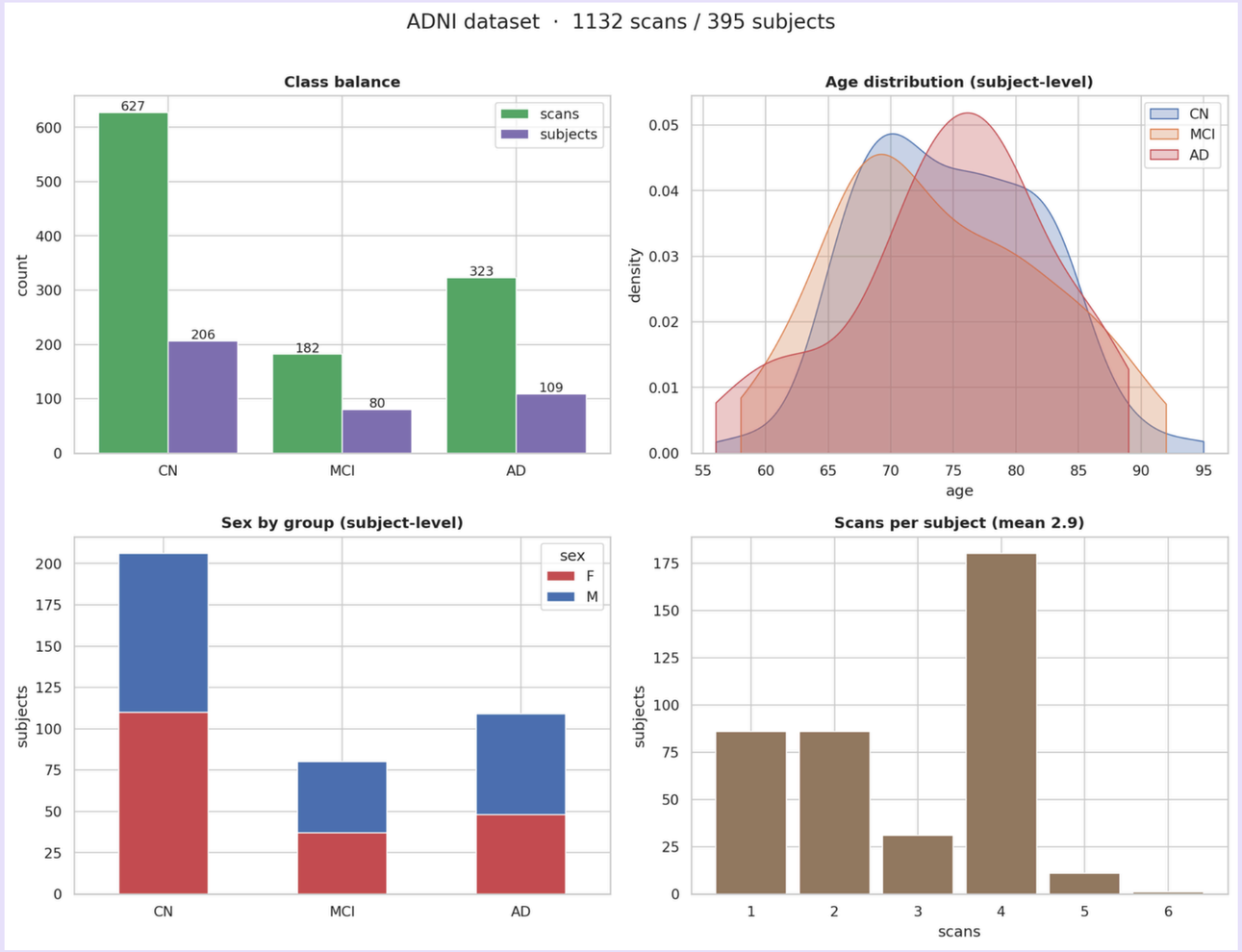
DATASET: ADNI

- ADNI GO/2/3 · 3T T1-WEIGHTED MPRAGE · PRE-PROCESSED NIFTI
- Comprises of 1,132 scans / 395 subjects
- Per-subject: CN-206, MCI-80, AD-109 (imbalanced; MCI smallest)
- 18 non-standard scans dropped — no subject lost
- Longitudinal scans require subject-level aggregation



CN vs AD — visible atrophy

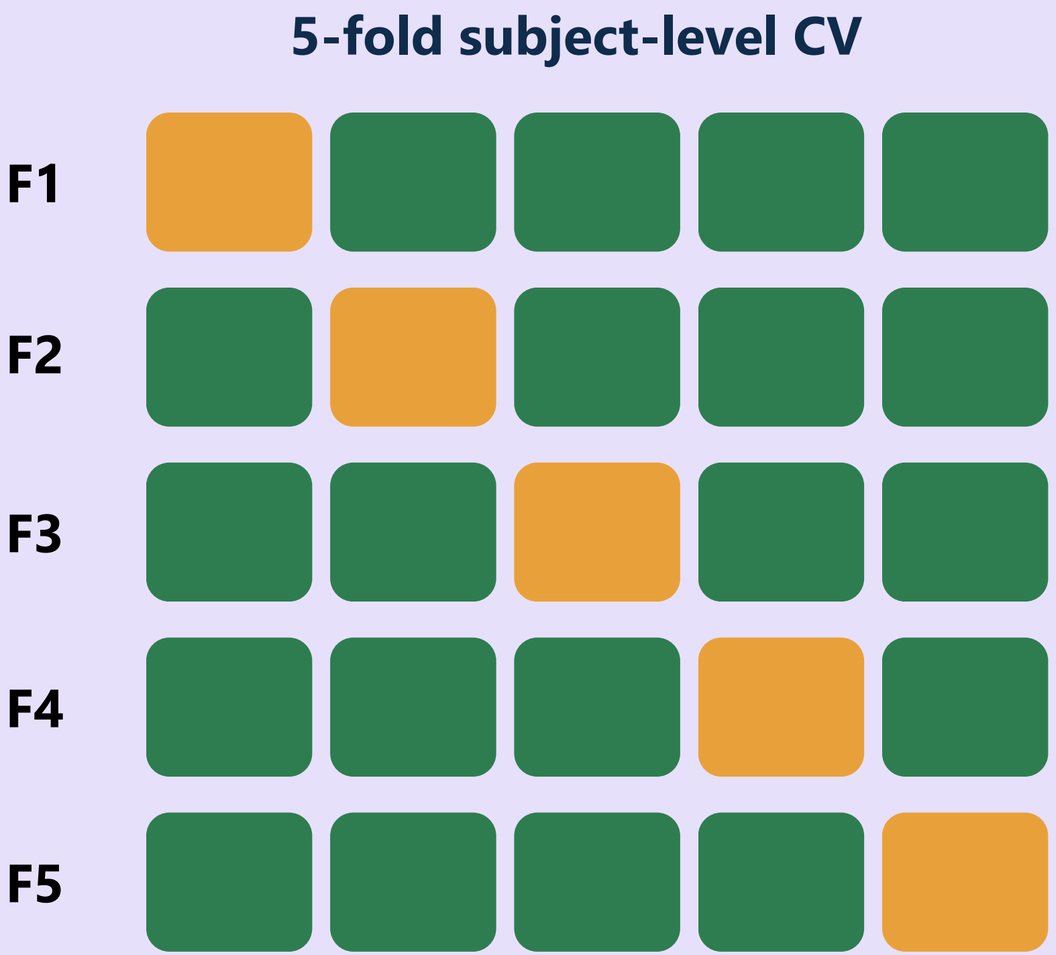
DATASET: ADNI



Class balance, age/sex, scans per subject

EVALUATION PROTOCOL

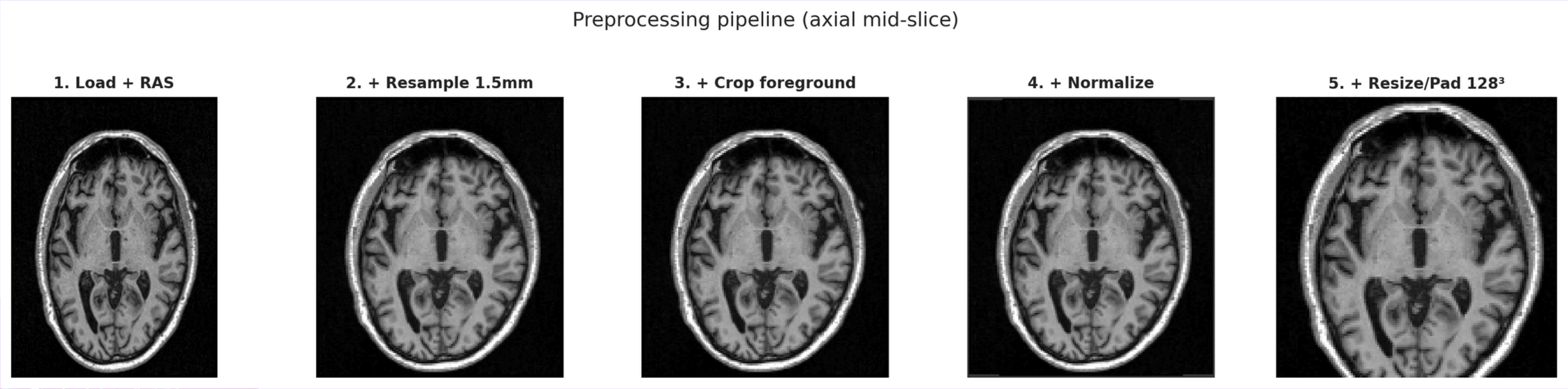
- Subject-level split: no patient in both train/test
- 5-fold cross-validation, seed=42
- Same folds for all models
- Early stopping using validation Macro-F1
- One final prediction per subject



Each subject is tested once across 5 runs to prevent leakage

PREPROCESSING

- Reorient to RAS
- Resample to 1.5 mm isotropic
- Foreground crop
- Per-volume z-score normalization
- Resize/pad to 128^3
- Persistent cache for faster training

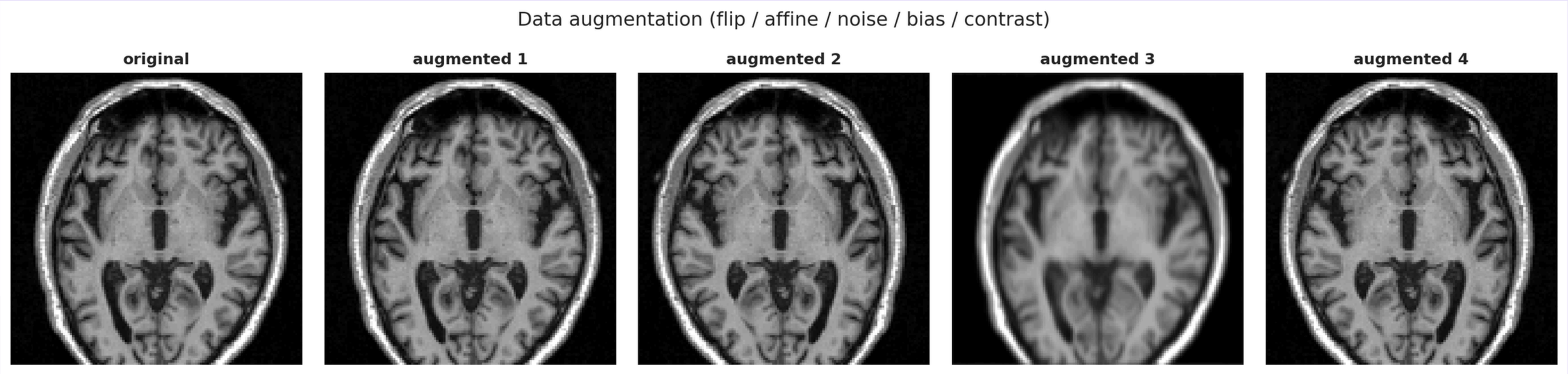


Preprocessing pipeline

PREPROCESSING

Train-only augmentation

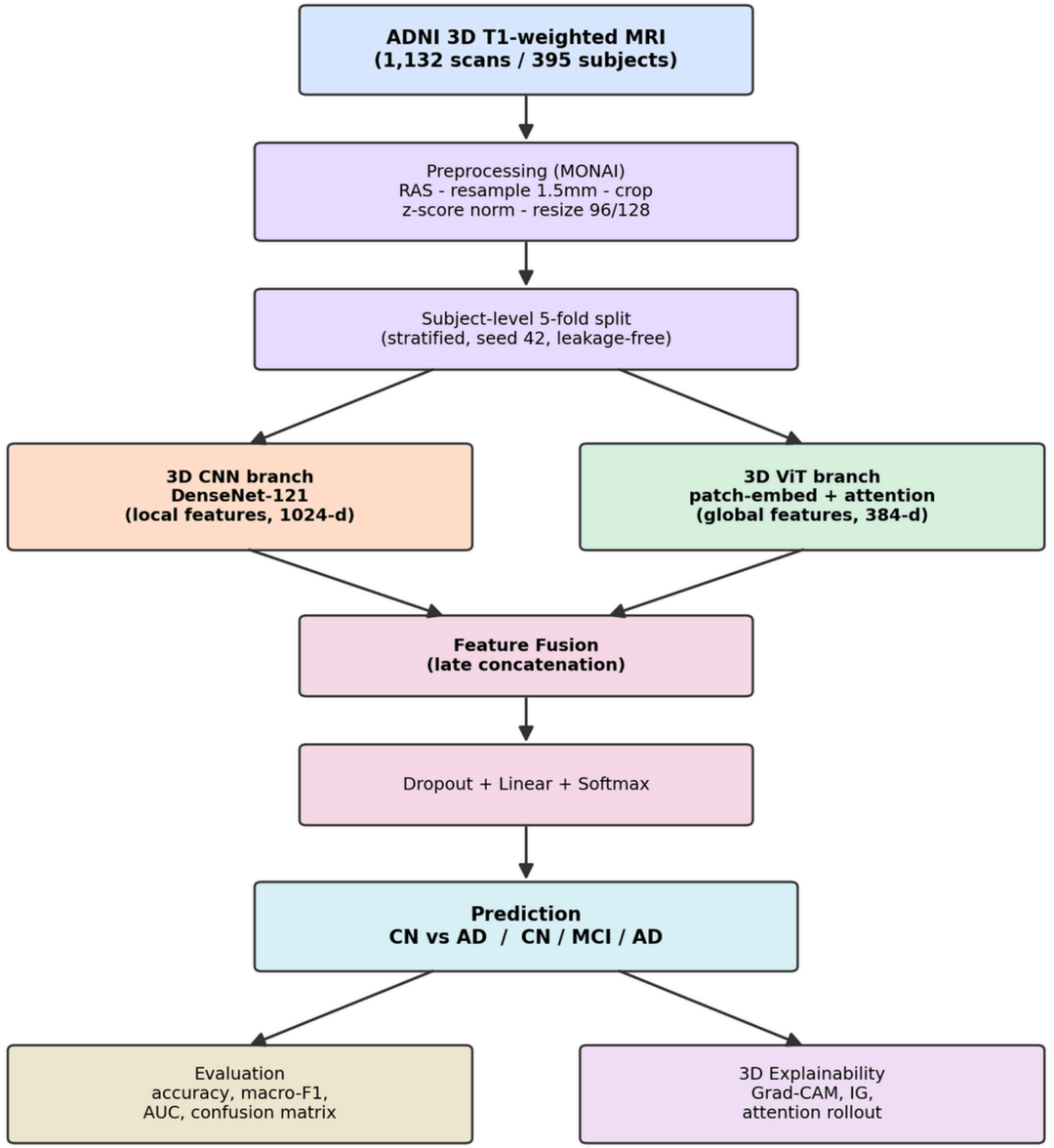
- Flip, affine, noise
- Bias field, intensity
- Mild elastic deformation
- 3D MixUp



Augmentation examples

11 3D ARCHITECTURES EVALUATED

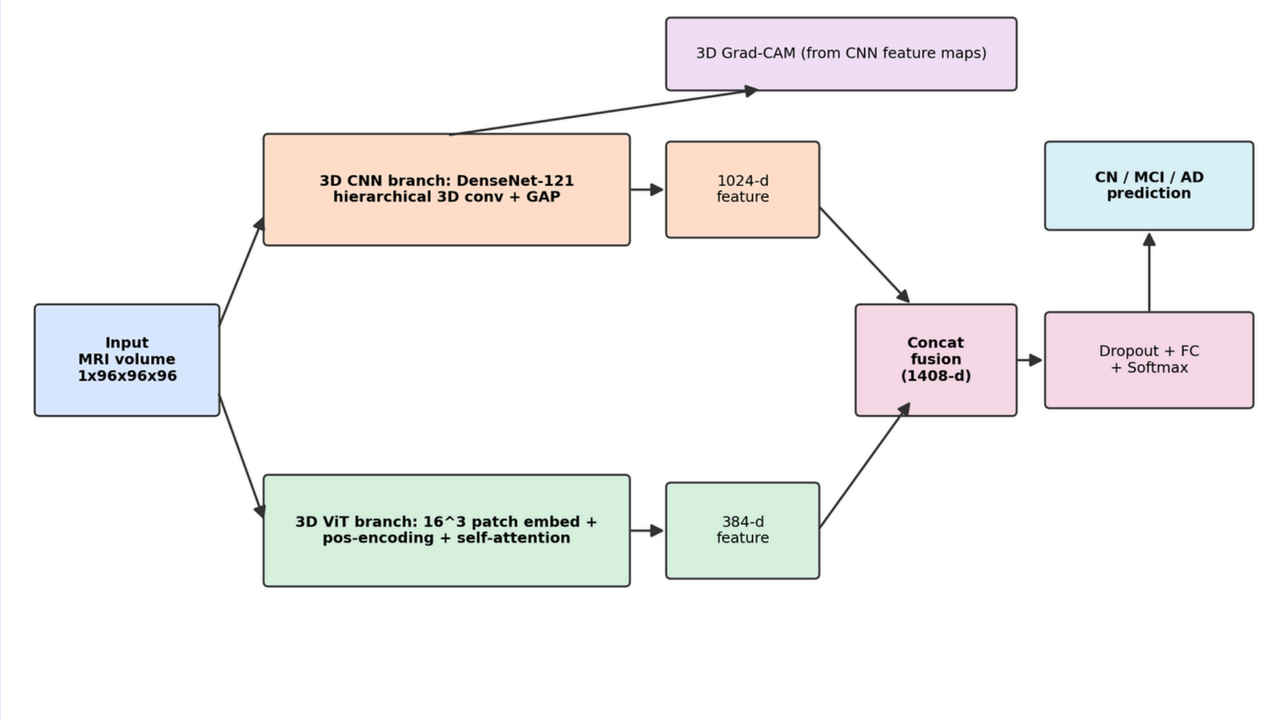
- 3D DenseNet-121 - CNN baseline
- MedicalNet 3D ResNet - pretrained medical model
- 3D ViT - transformer from scratch
- 3D Swin Transformer - hierarchical transformer
- Hybrid CNN-ViT/CNN-Swin- feature fusion



Five backbones and one shared evaluation

HYBRID CNN + TRANSFORMER

- CNN branch learns local 3D anatomical patterns
- Transformer branch models global brain context
- Feature fusion combines local + global representations
- Concatenated features go to a shared classifier head
- Same architecture supporting both CN vs AD and CN / MCI / AD



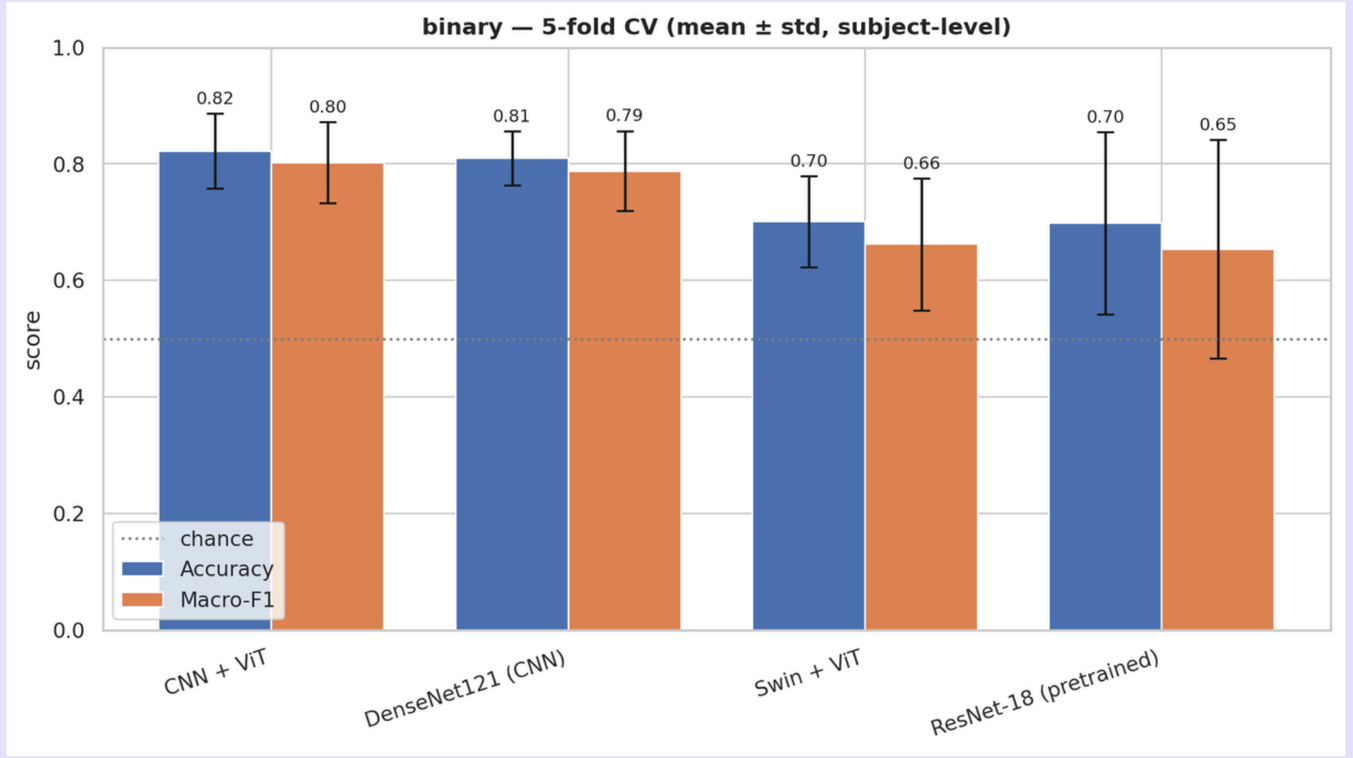
Local CNN features + global Transformer features fused through concatenation for final classification

RESULTS: BINARY CN VS AD

82.2%

Hybrid 3D CNN–ViT · leakage-free

Model	Accuracy	Macro-F1	
Hybrid CNN–ViT (best)	82.2 ± 5.7	0.802 ± 0.062	
DenseNet-121 (CNN)	81.0 ± 4.1	0.788 ± 0.062	
Swin–ViT (no CNN)	70.2 ± 7.0	0.662 ± 0.101	
MedicalNet ResNet	69.8 ± 13.9	0.654 ± 0.168	



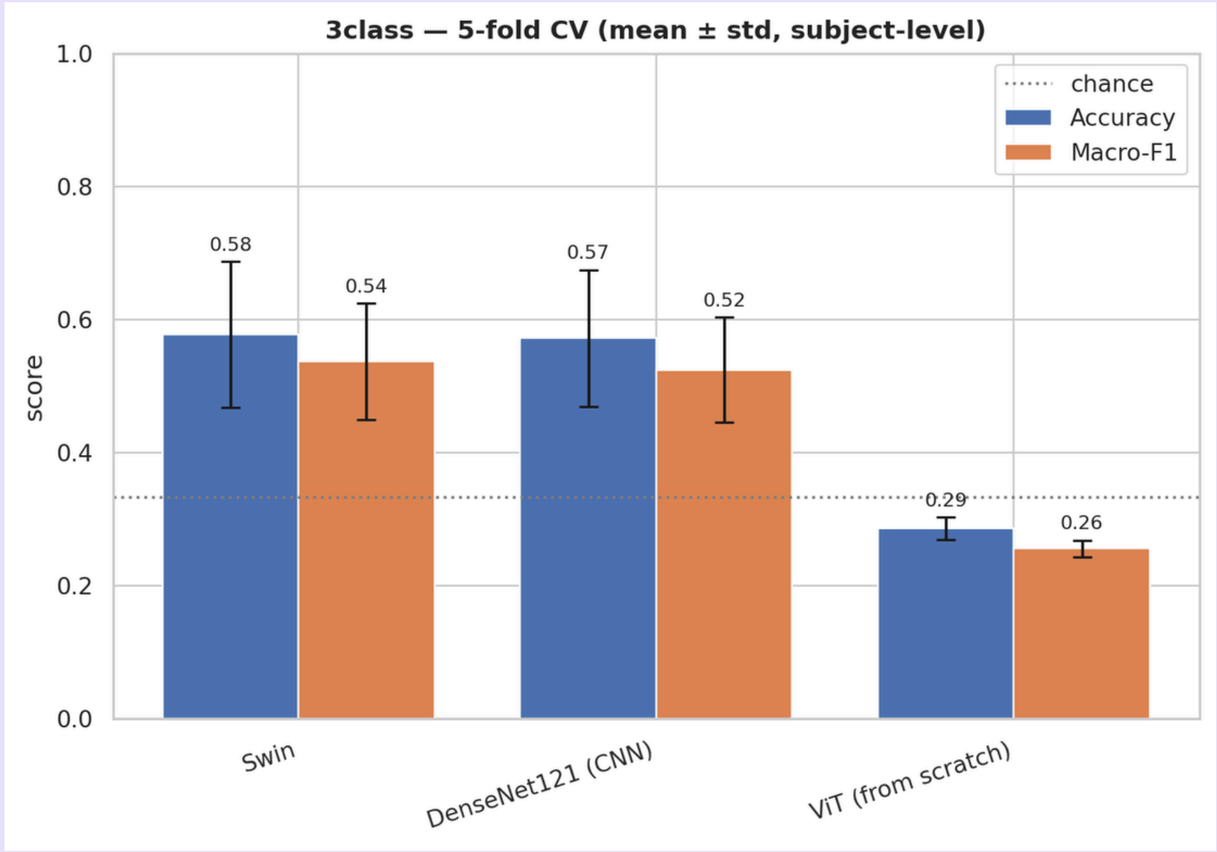
Accuracy & Macro-F1 (mean ± std)

RESULTS: 3-CLASS CN / MCI / AD

57.7%

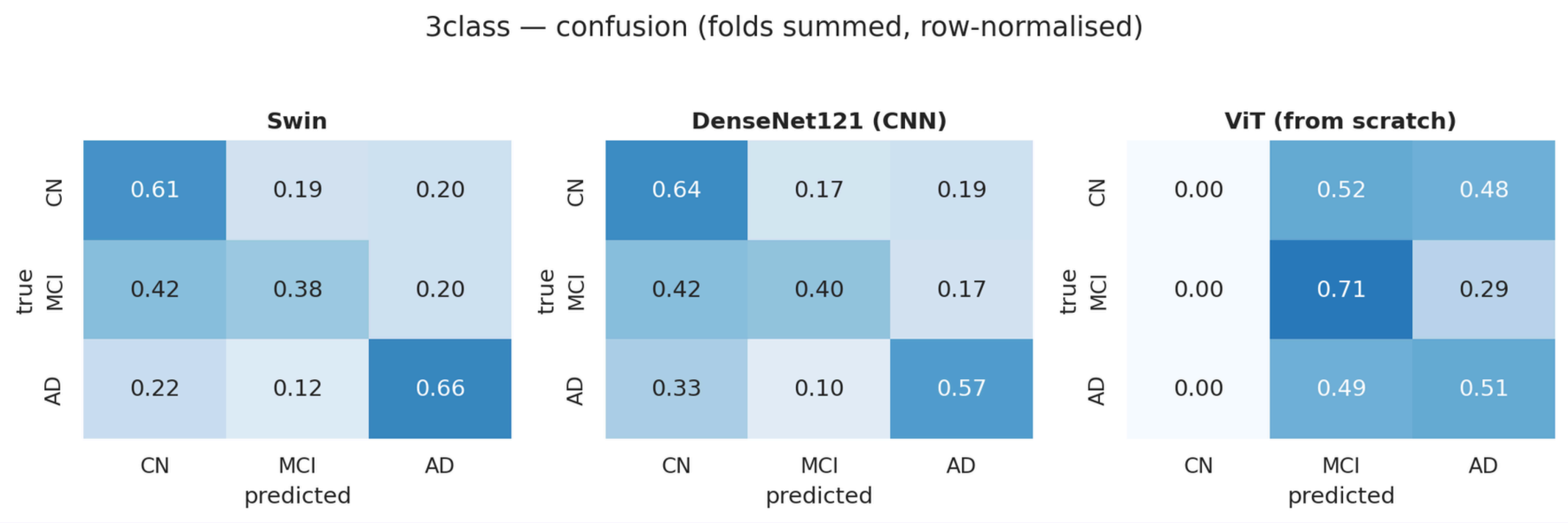
Swin · MCI is the bottleneck

Model	Accuracy	Macro-F1
Swin Transformer (best)	57.7 ± 9.8	0.537 ± 0.078
DenseNet-121 (CNN)	57.2 ± 9.2	0.524 ± 0.071
3D ViT (scratch)	28.6 ± 1.5	0.256 ± 0.012



Per-model 3-class

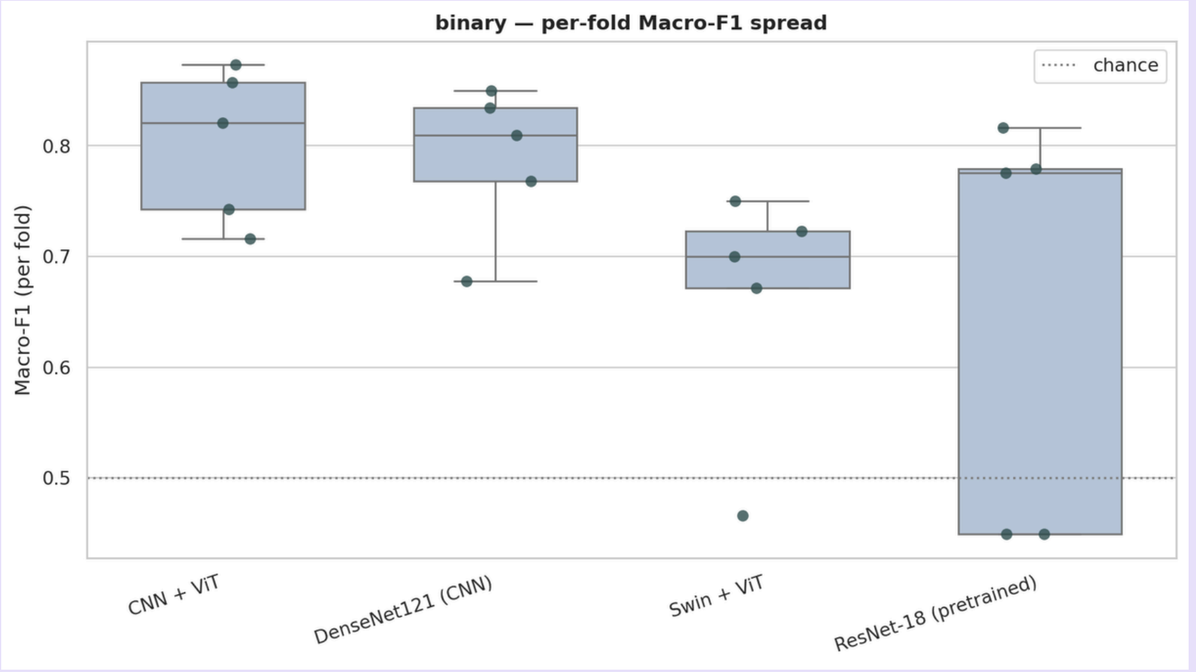
RESULTS: 3-CLASS CN / MCI / AD



Confusion Matrix for 3-Class Alzheimer's Classification (CN, MCI, AD)

RQ1 ANSWER: HYBRIDS HELP — BUT MODESTLY

- Hybrid CNN–ViT achieved the best performance, with only a slight improvement over DenseNet.
- Performance depends on the fusion strategy used.
- CNN is the most important component; removing it significantly reduces Macro-F1.
- Transformer-only models are the weakest and least stable

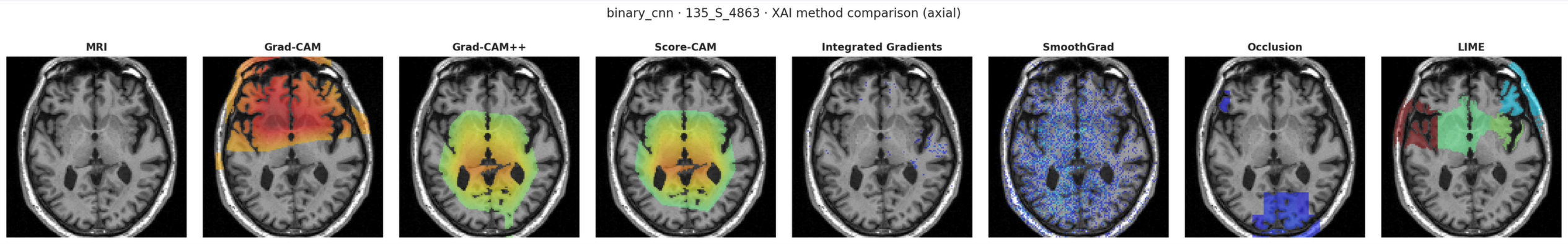


EXPLAINABILITY (RQ2): ARE THE XAI MAPS ANATOMICALLY PLAUSIBLE?

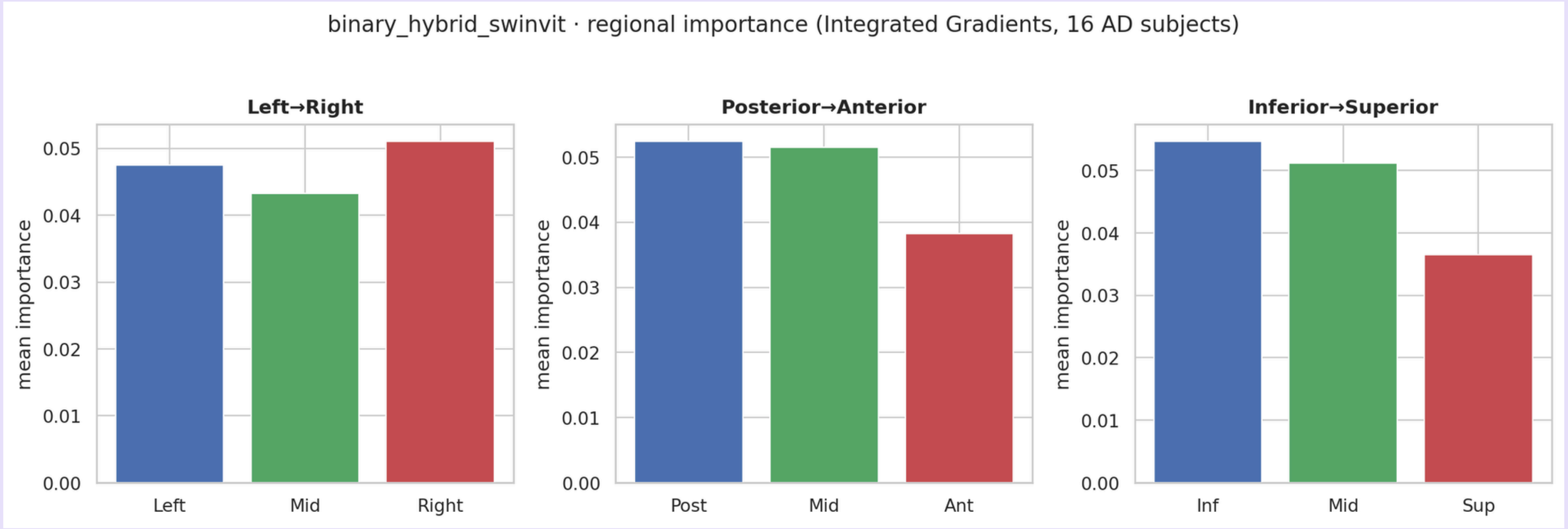
3D attribution suite: Grad-CAM, Grad-CAM++, Integrated Gradients, SmoothGrad, Occlusion, LIME

Importance concentrates in medial-temporal & peri-ventricular regions

Agreement across methods AND across models (CNN, Swin, hybrid)



EXPLAINABILITY (RQ2): DOES IT SEE AD BIOLOGY?

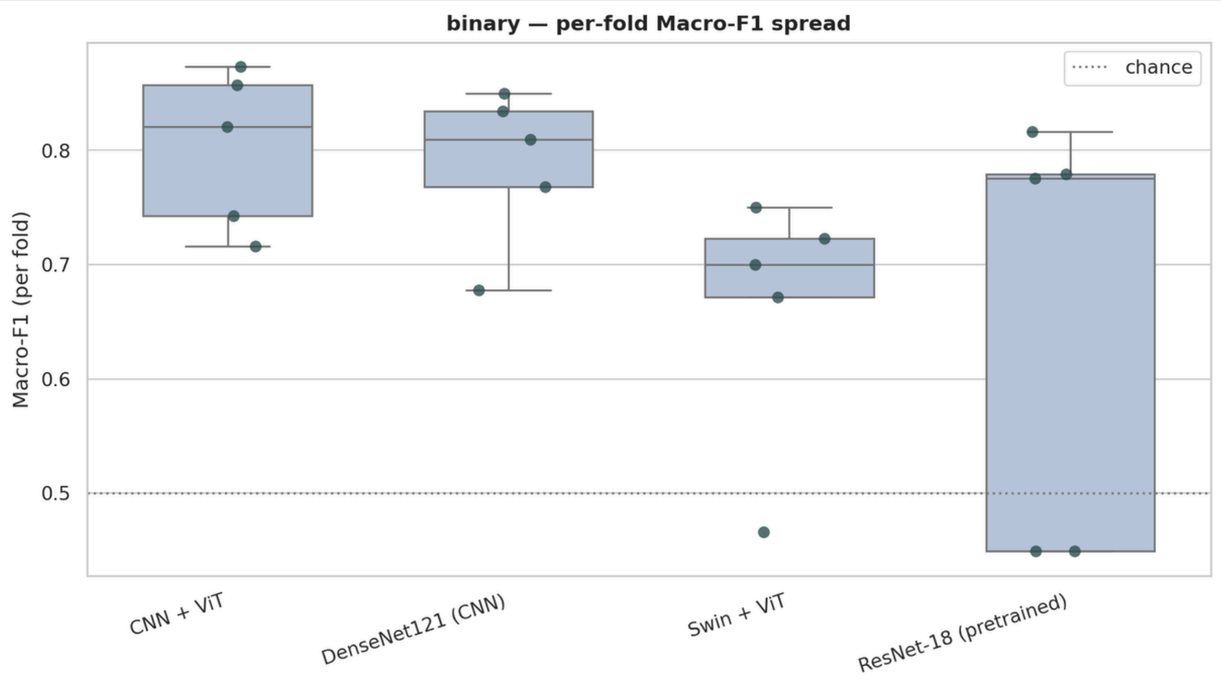


PER-FOLD MACRO-F1

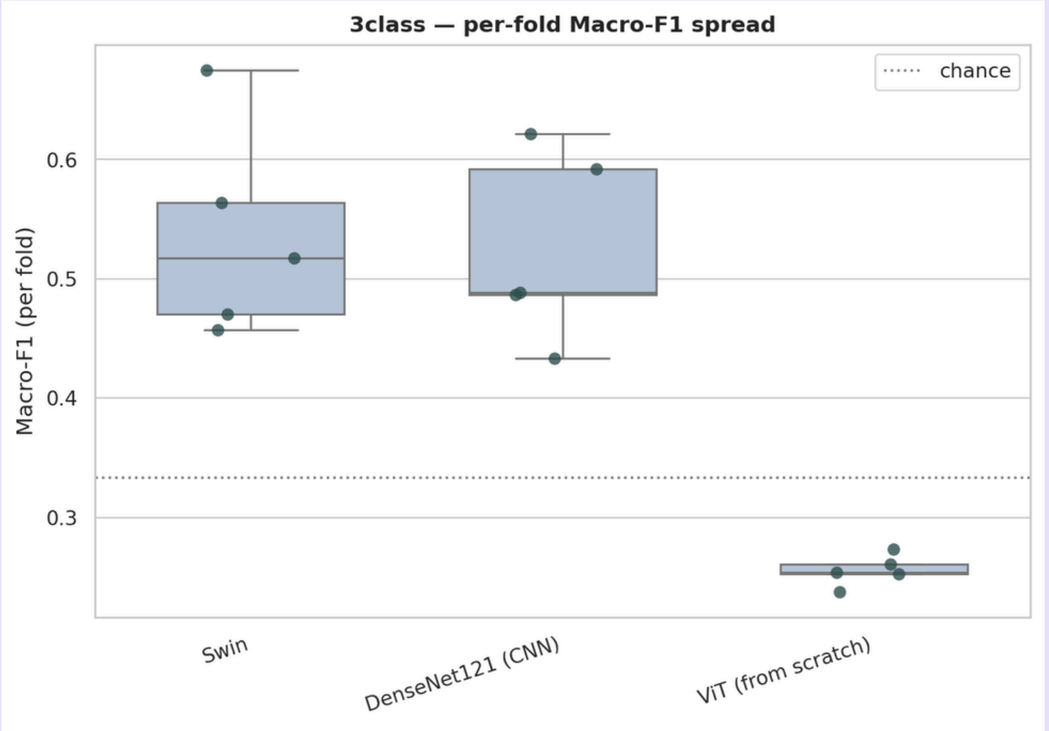
Binary CNN–ViT folds: 0.716 / 0.857 / 0.743 / 0.821 / 0.873

Binary ResNet folds: 0.775 / 0.449 / 0.449 / 0.779 / 0.816

Mean ± std hides this spread - hence per-fold reporting

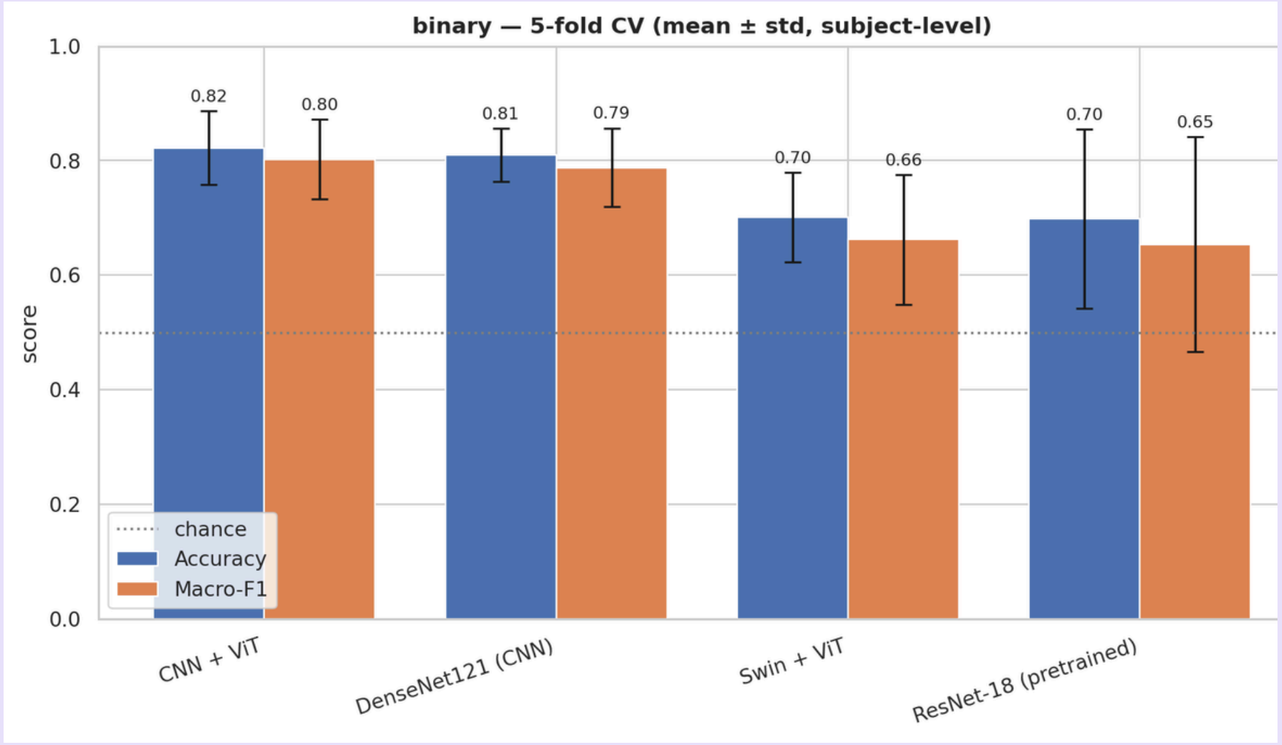


Binary — per-fold

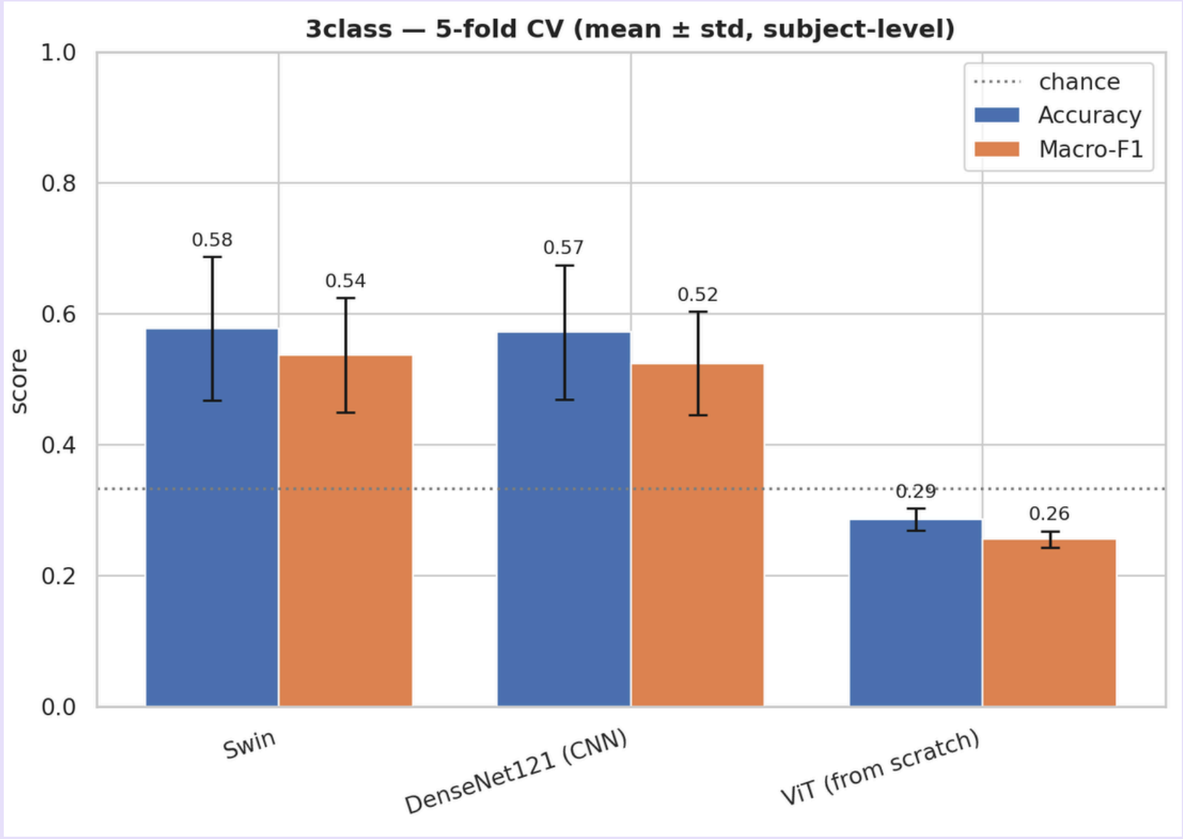


3-class — per-fold

SUBJECT-LEVEL CV RESULTS



Binary CN vs AD Performance Across 5 Folds



Three-Class CN/MCI/AD Performance Across 5 Folds

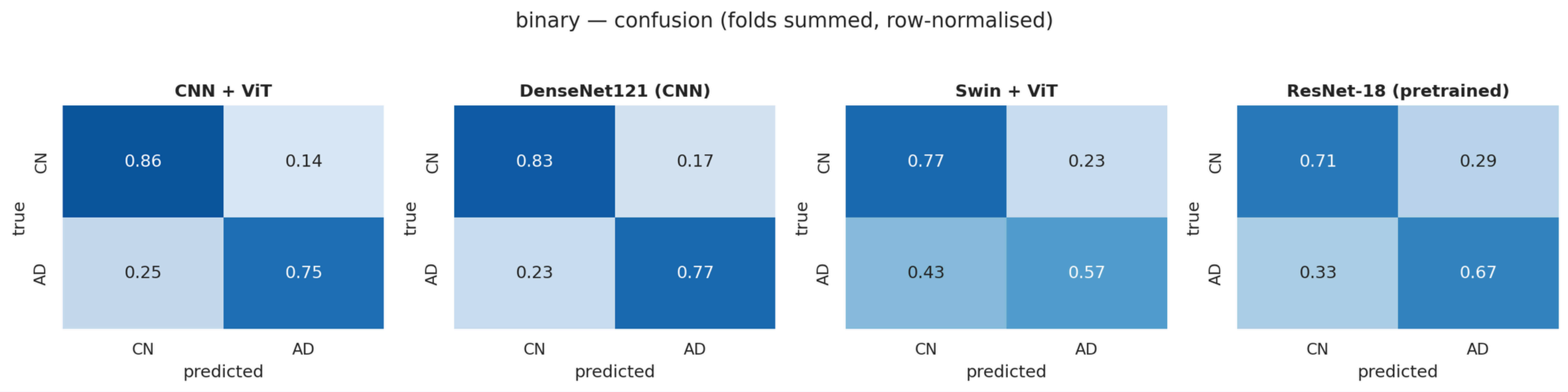
PER-CLASS PRECISION / RECALL / F1

	Precision	Recall	F1
Binary — CN	0.87	0.86	0.86
Binary — AD	0.76	0.74	0.75
3-class — CN	~0.66	~0.65	~0.65
3-class — MCI	~0.39	~0.38	0.37–0.41
3-class — AD	~0.58	~0.60	~0.55–0.61

Class-wise Performance Metrics

Precision, recall, and F1-score for binary and 3-class classification; MCI shows the lowest performance due to overlap with CN and AD.

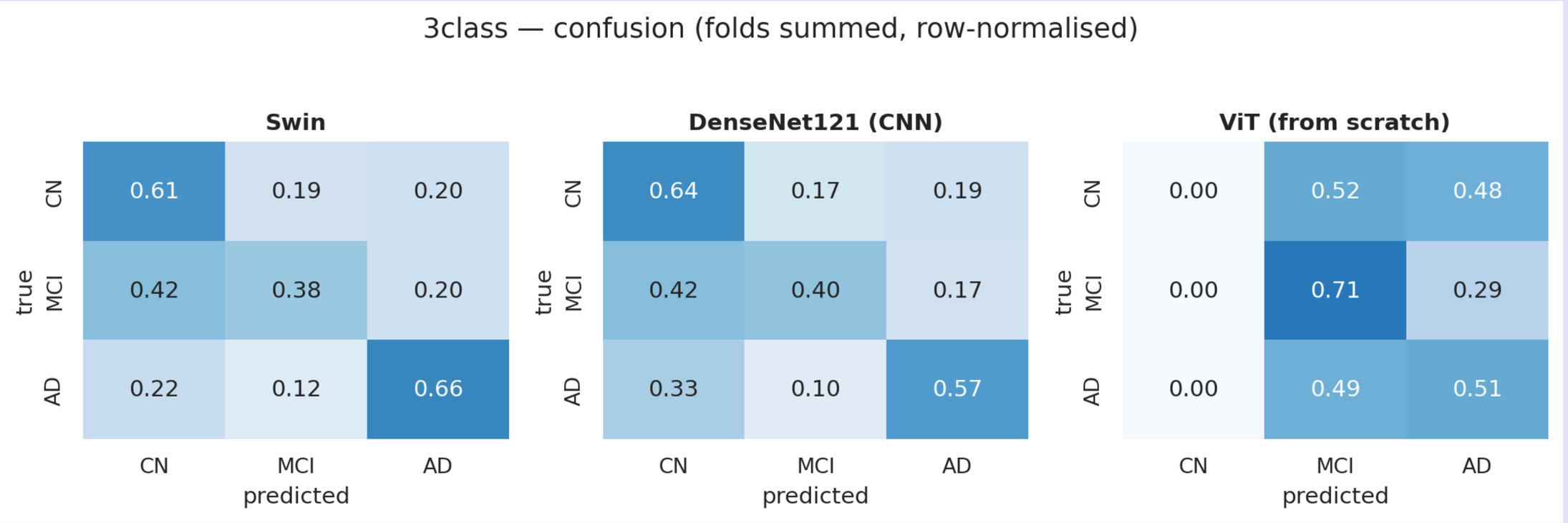
BINARY ERROR ANALYSIS



CN vs AD Confusion Matrices

Fold-summed, row-normalized confusion matrices showing strong separation between CN and AD.

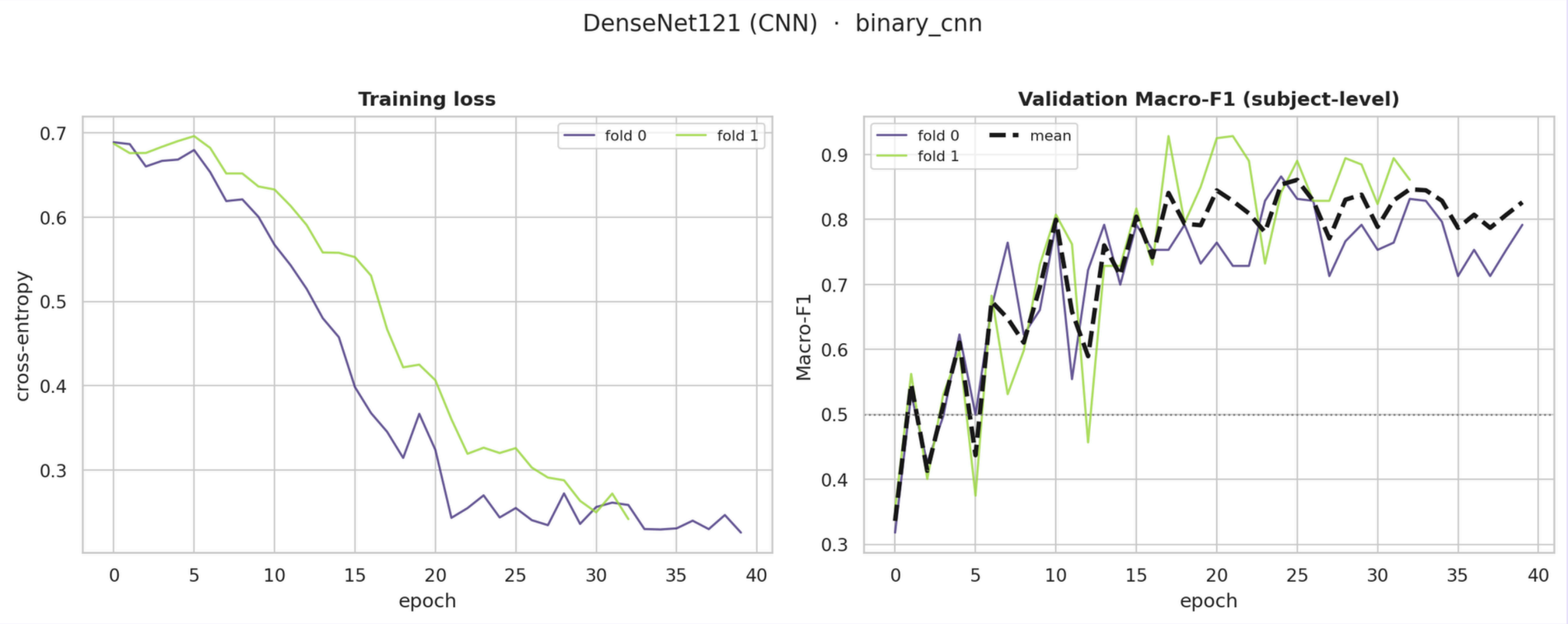
3-CLASS ERROR ANALYSIS



CN/MCI/AD Confusion Matrices

Fold-summed, row-normalized confusion matrices showing that MCI is frequently confused with CN and AD.

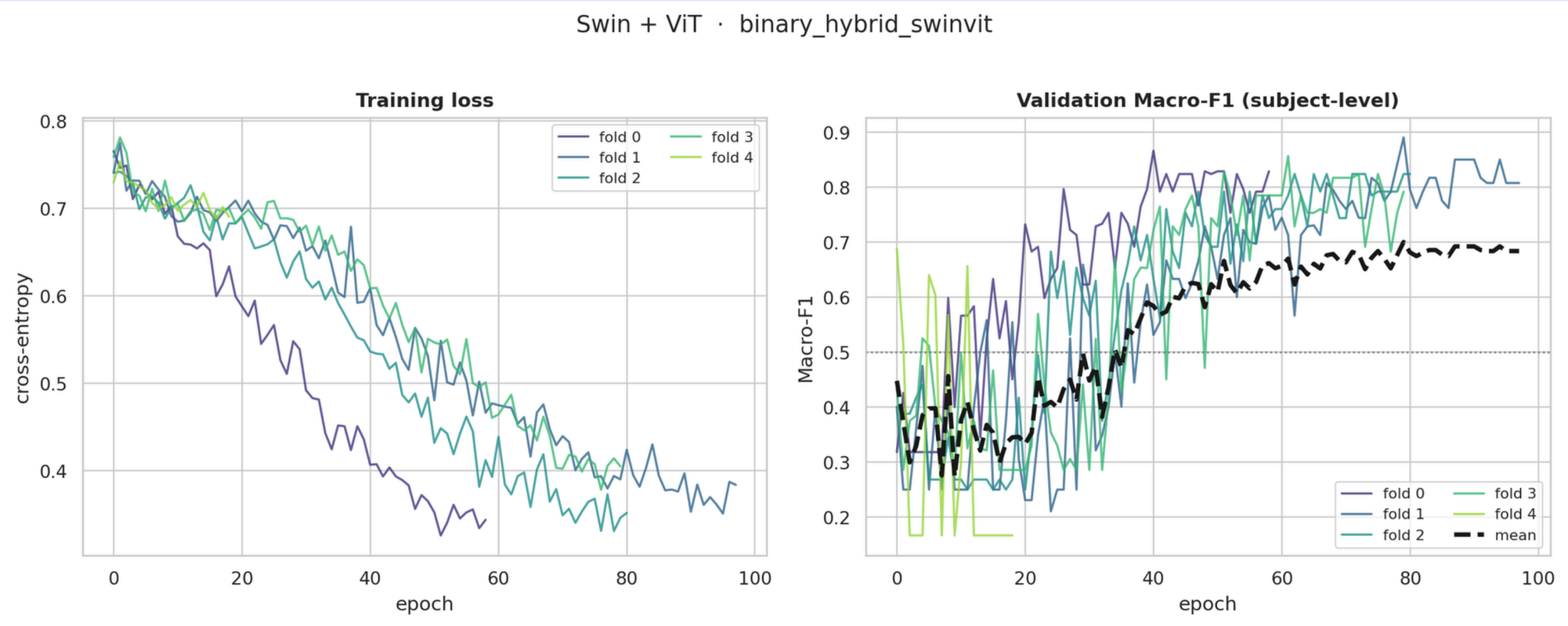
TRAINING DYNAMICS



DenseNet121 Binary Training and Validation Curves

DenseNet121 shows smooth loss reduction and strong validation Macro-F1, confirming that a 3D CNN is a reliable baseline for binary CN vs AD classification.

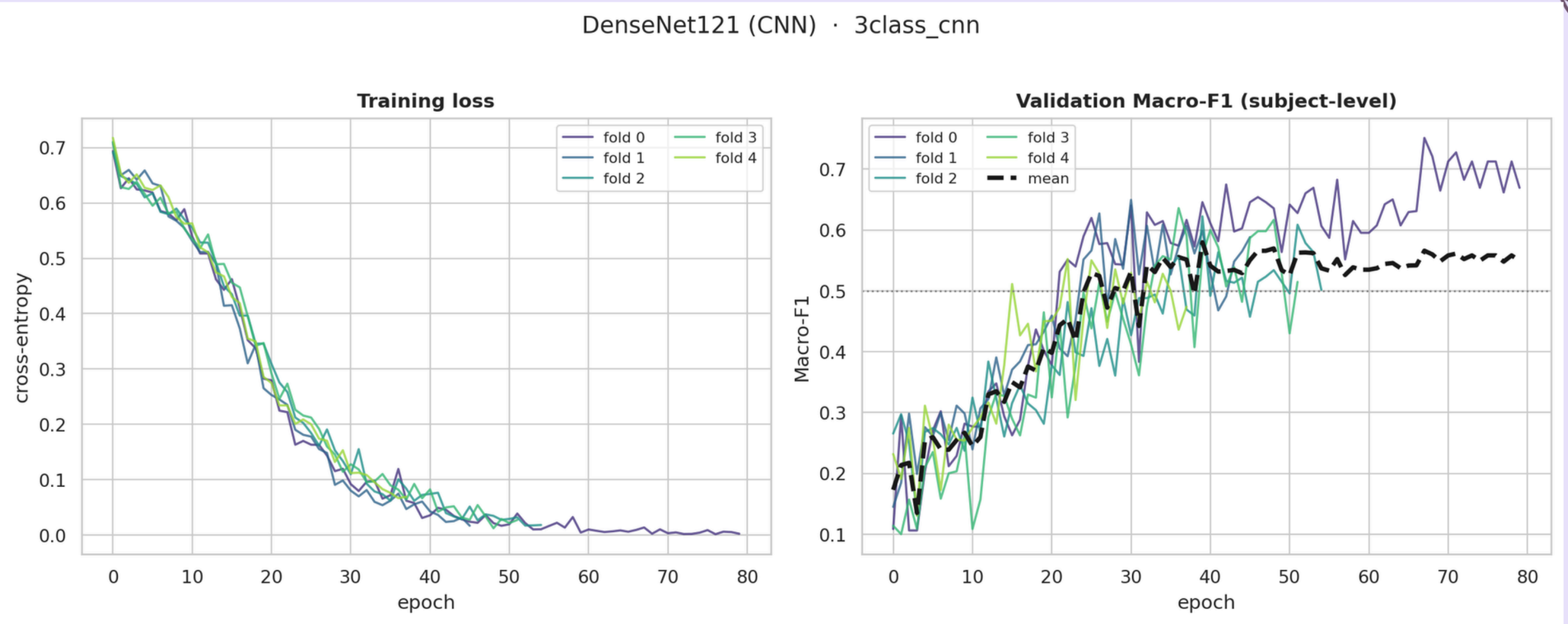
TRAINING DYNAMICS



Swin + ViT Binary Training and Validation Curves

Swin + ViT improves validation Macro-F1, but fold-wise variability indicates unstable generalization.

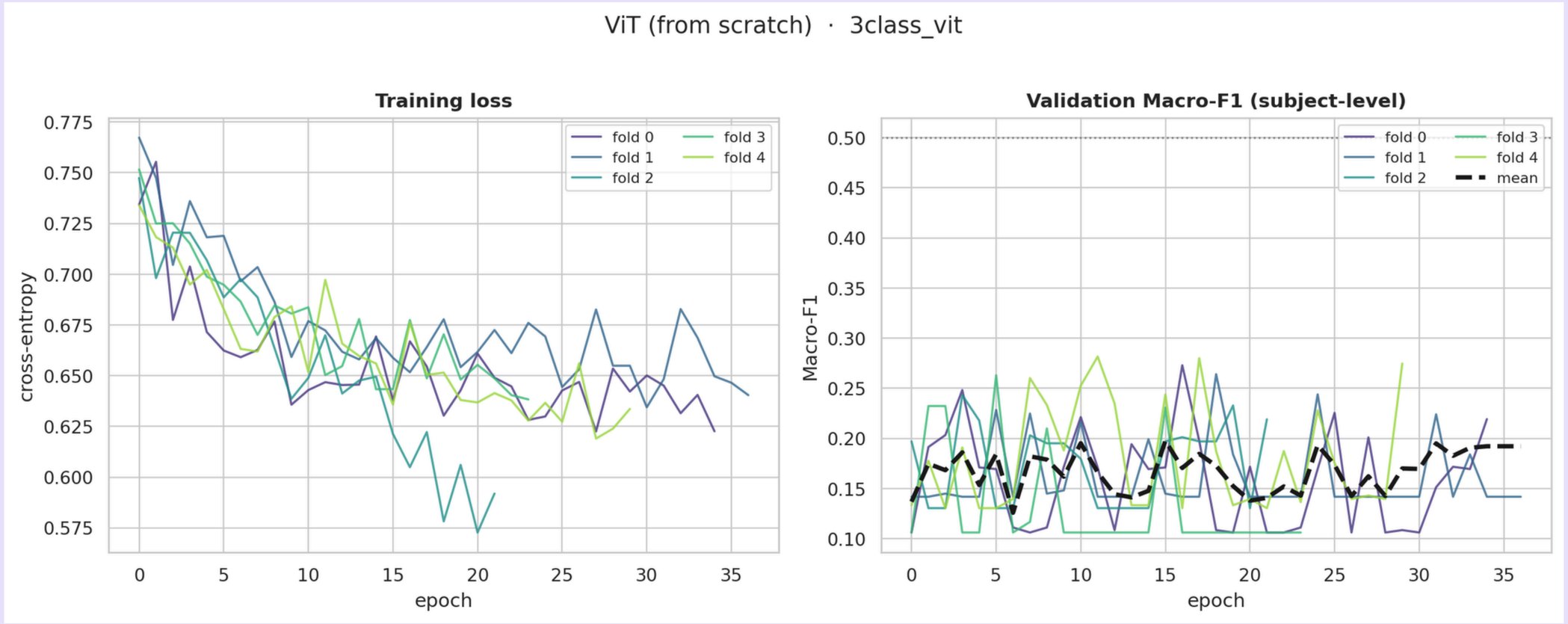
TRAINING DYNAMICS



DenseNet121 3-Class Training and Validation Curves

DenseNet121 shows strong loss convergence and validation Macro-F1 improves above chance, but moderate performance reflects difficulty of separating MCI from CN and AD.

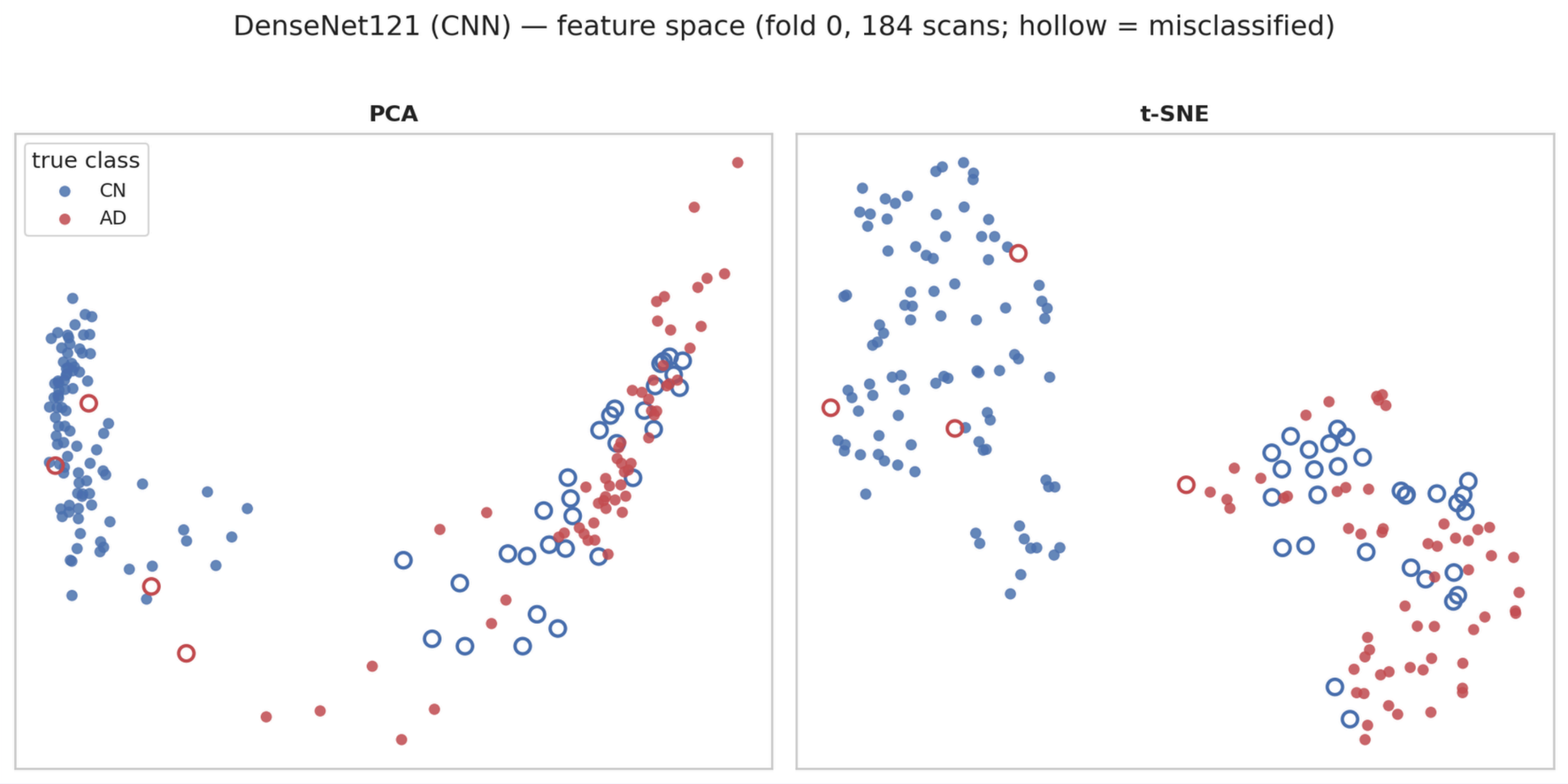
TRAINING DYNAMICS



ViT From Scratch: Training Loss and Validation Macro-F1

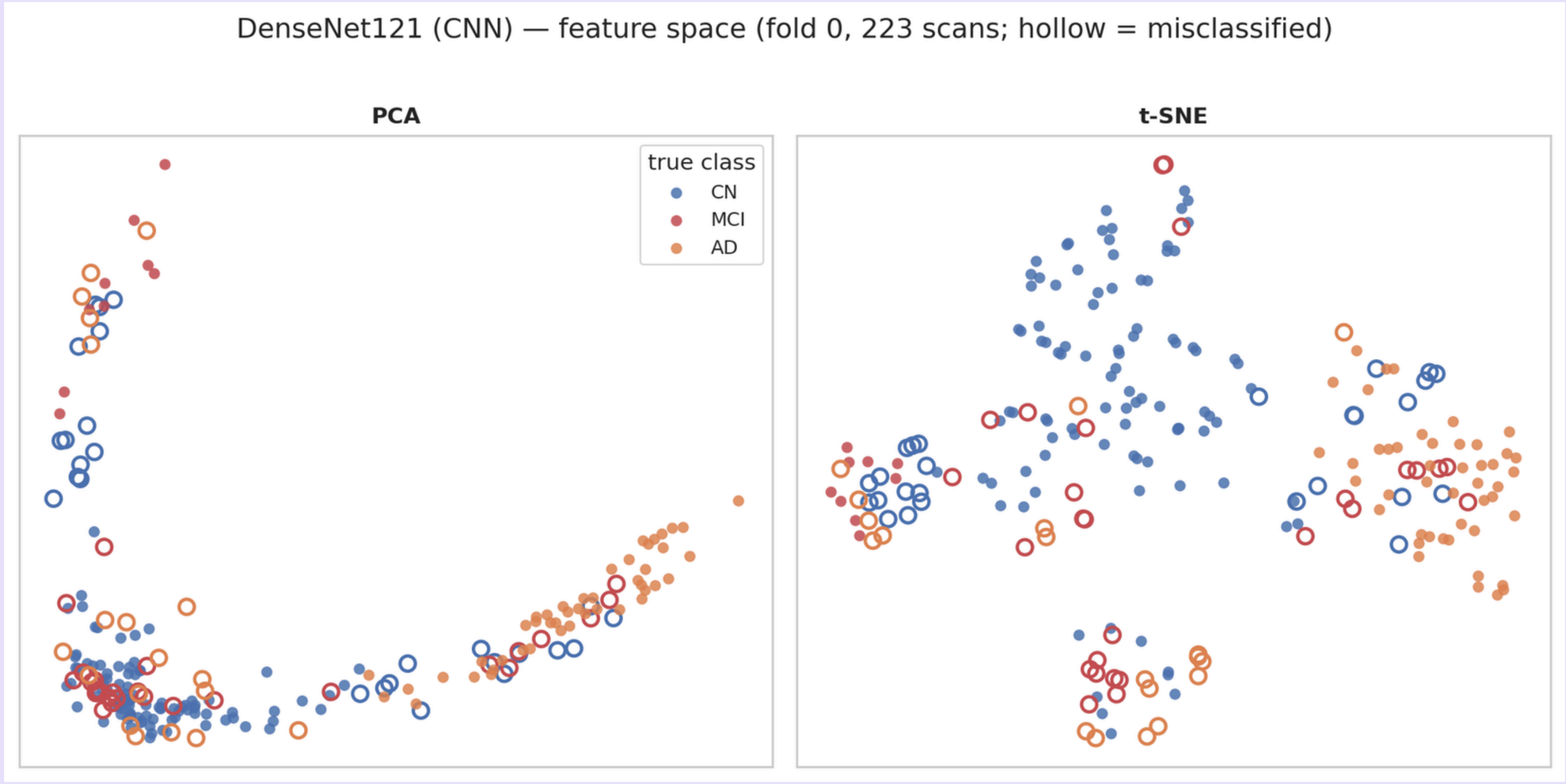
Loss decreases, but validation Macro-F1 stays near chance, showing ViT from scratch fails on 3-class learning.

FEATURE SPACE (T-SNE)



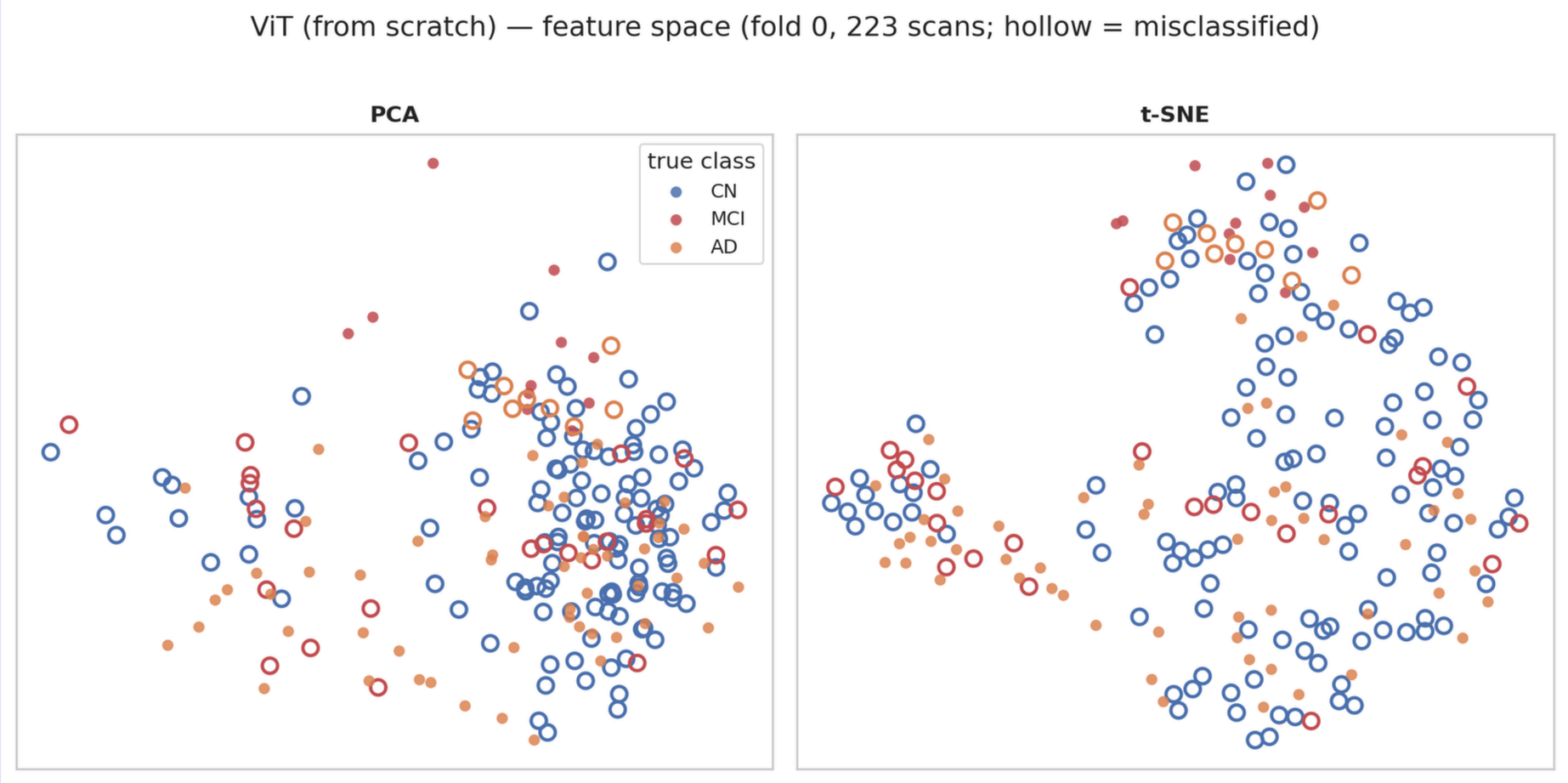
DenseNet121 Binary Feature Space

FEATURE SPACE (T-SNE)



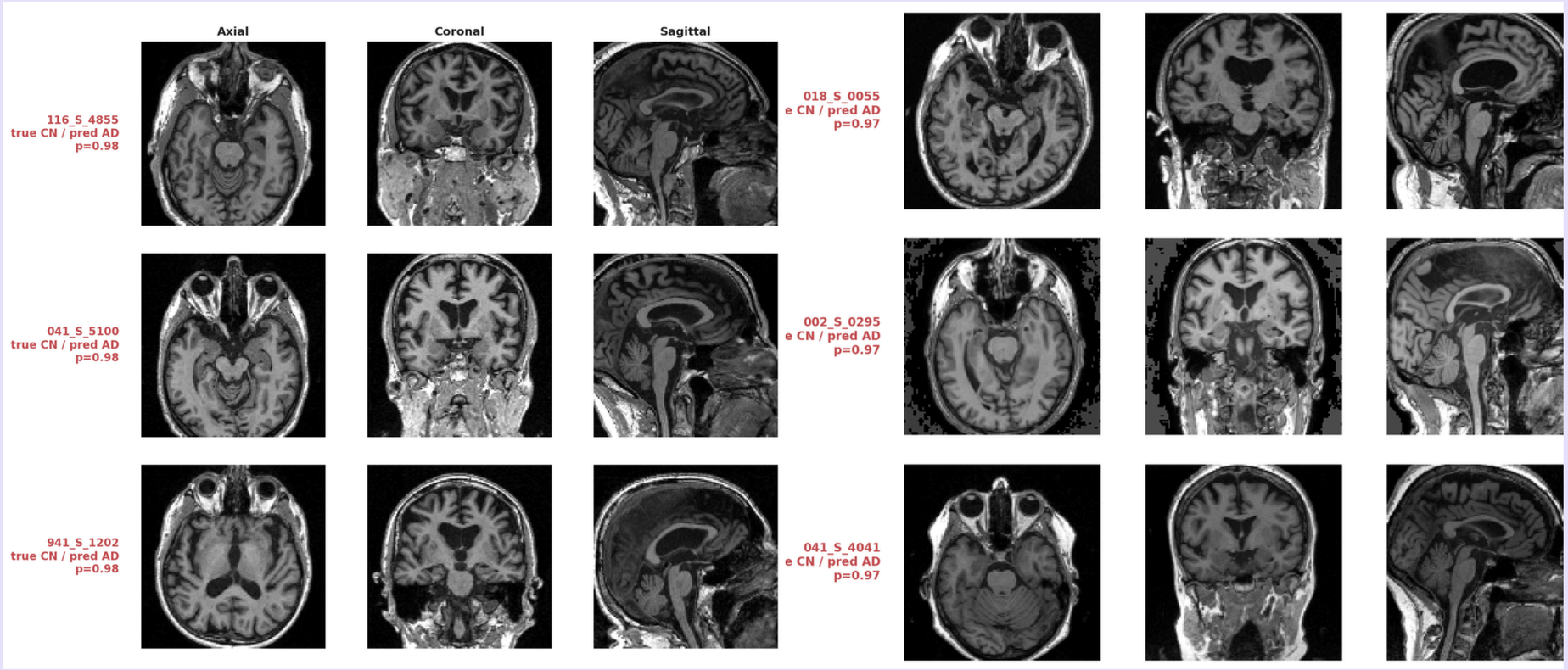
DenseNet121 CN/MCI/AD Feature Space

FEATURE SPACE (T-SNE)



ViT From Scratch: CN/MCI/AD Feature Space

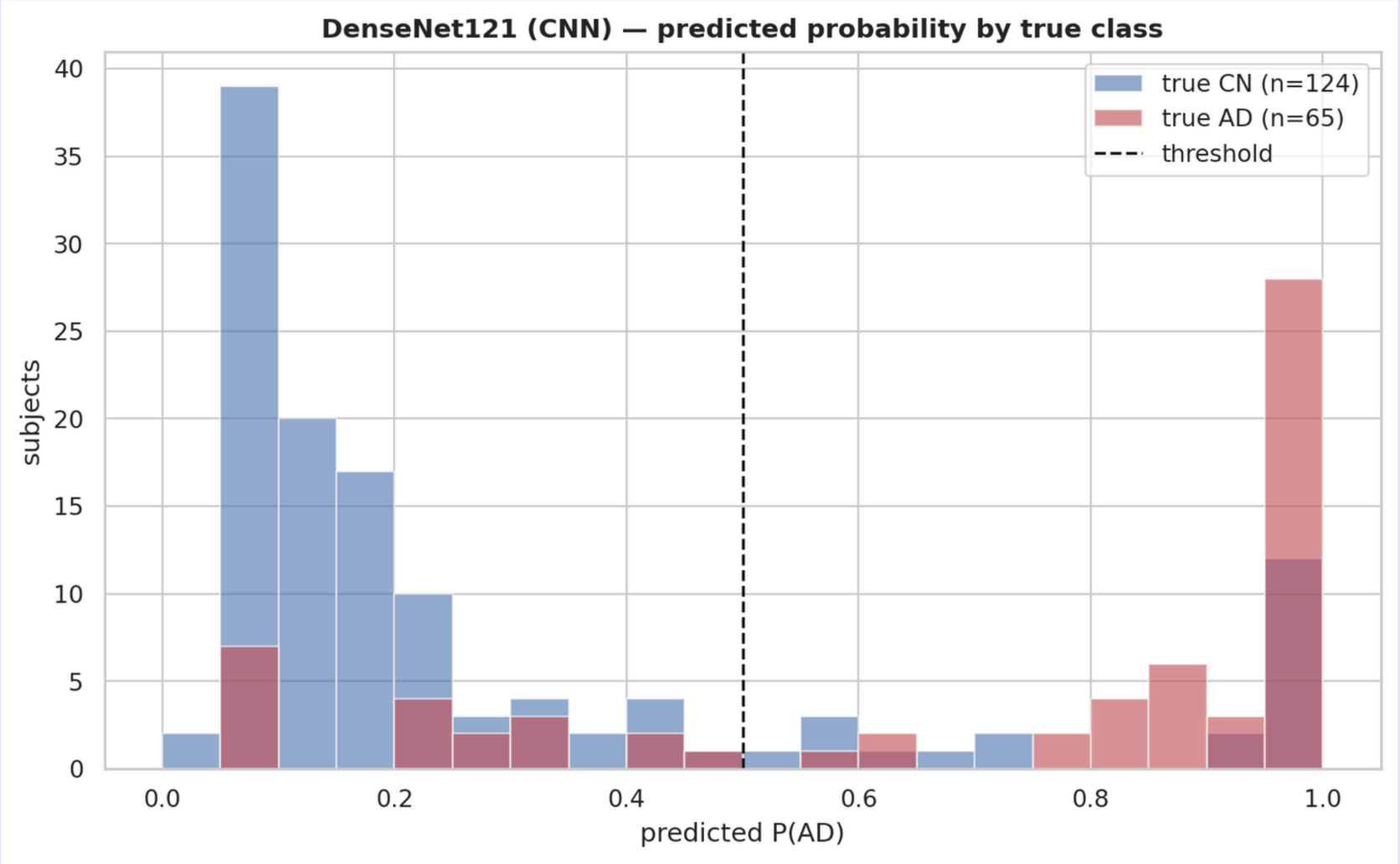
HIGH-CONFIDENCE ERROR CASES



DenseNet121 Most Confident Misclassifications

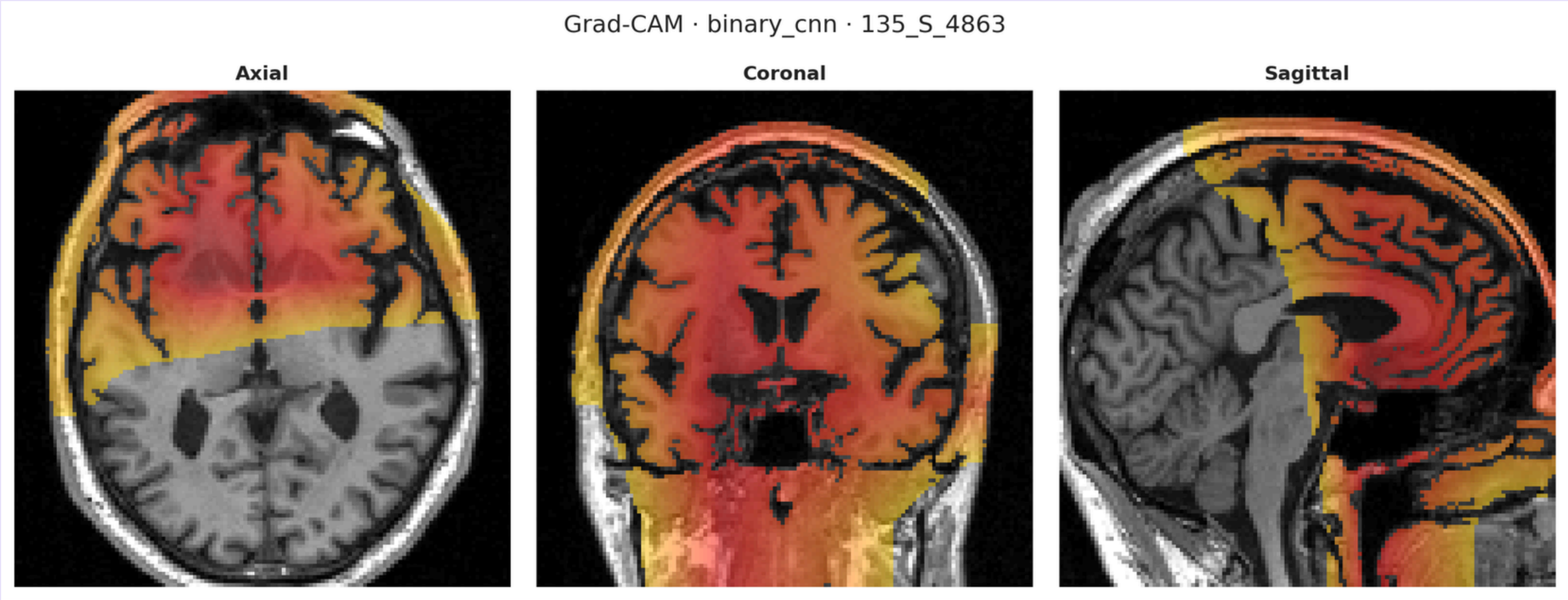
DenseNet121 occasionally predicts CN subjects as AD with very high confidence, showing that some normal cases contain AD-like structural patterns.

PREDICTION CONFIDENCE ANALYSIS



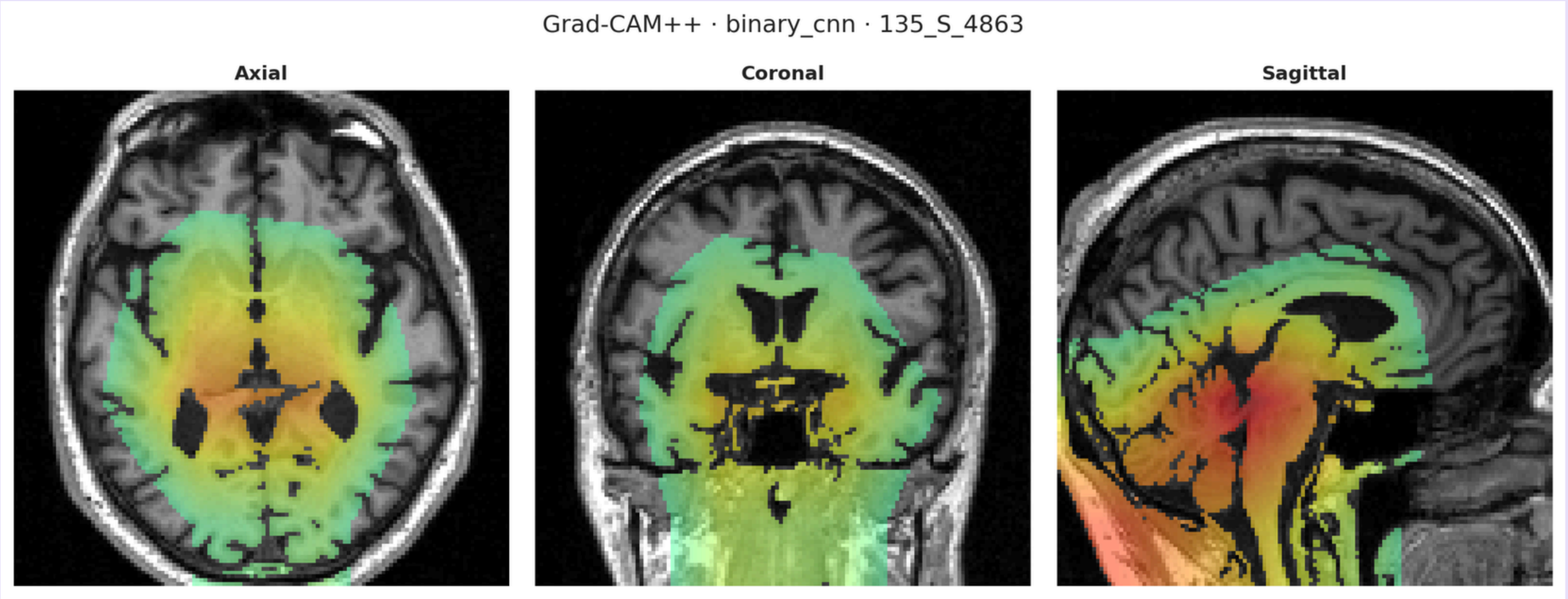
DenseNet121 Predicted AD Probability Distribution

XAI COMPARISON FOR BINARY CNN 33



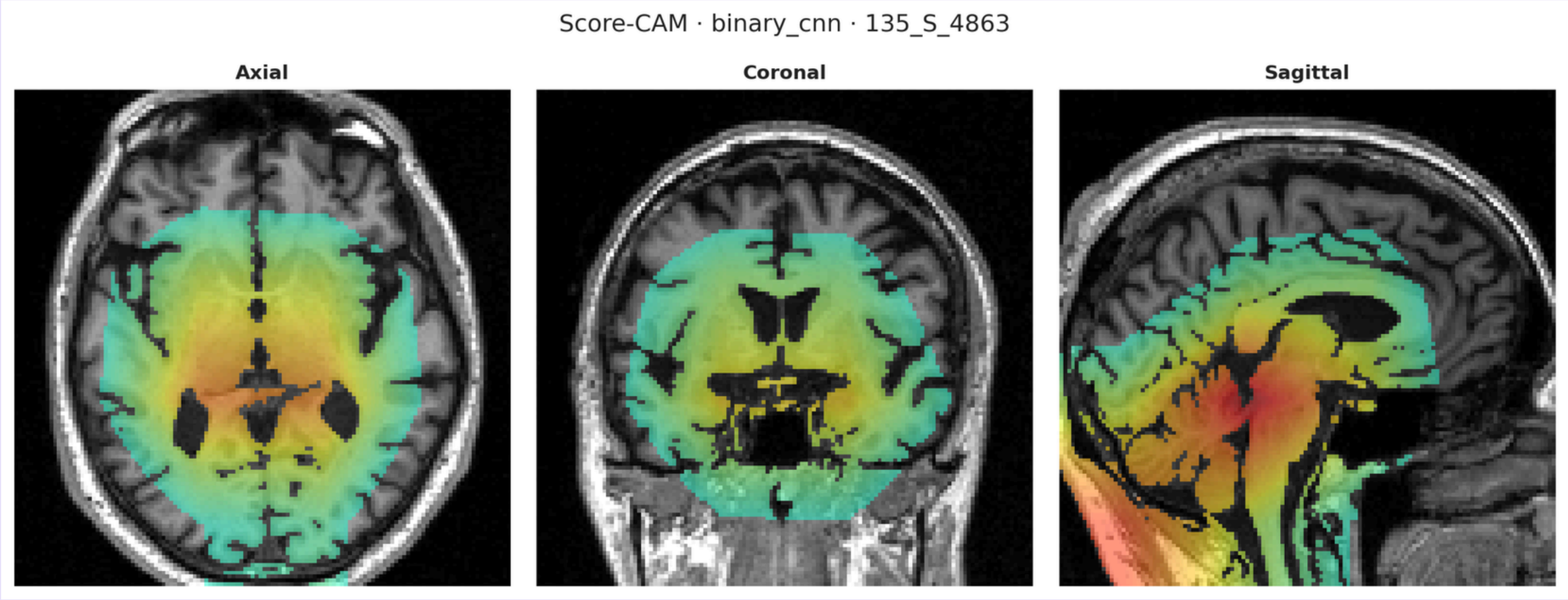
Grad-CAM Heatmap for DenseNet121 Binary CNN

XAI COMPARISON FOR BINARY CNN 34



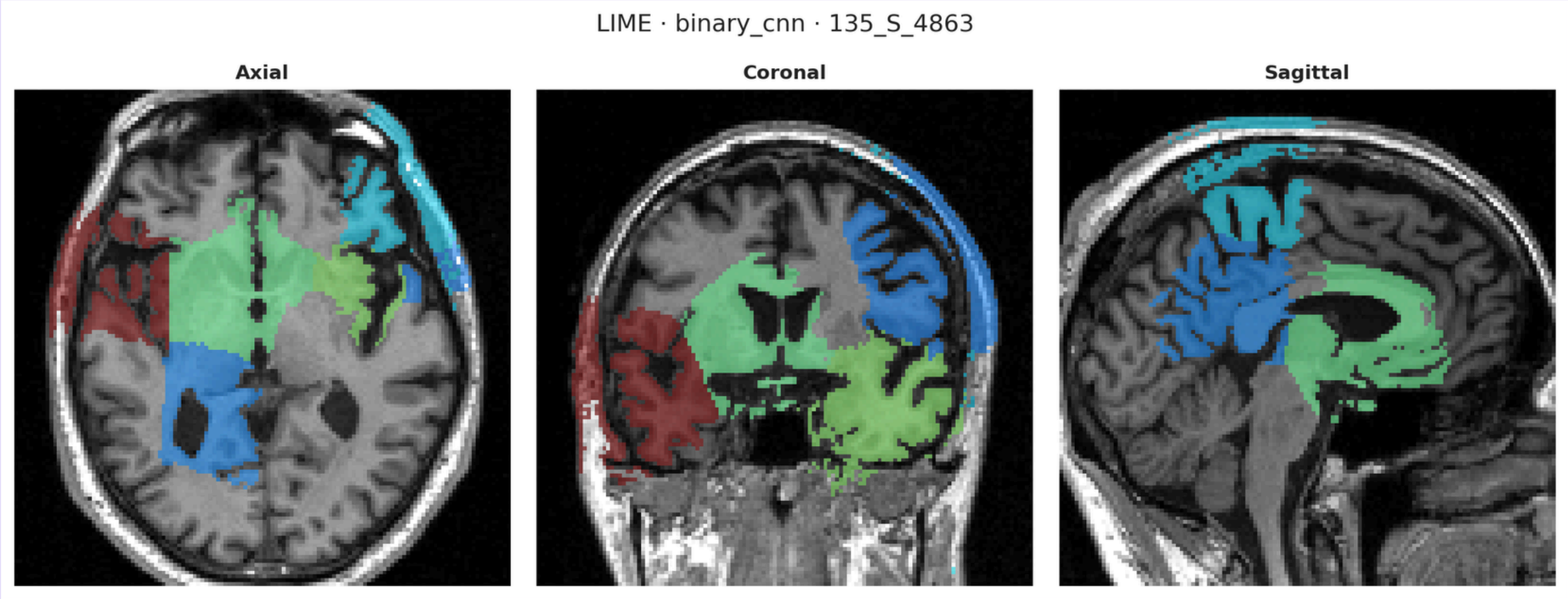
Grad-CAM++ Heatmap for DenseNet121 Binary CNN

XAI COMPARISON FOR BINARY CNN 35



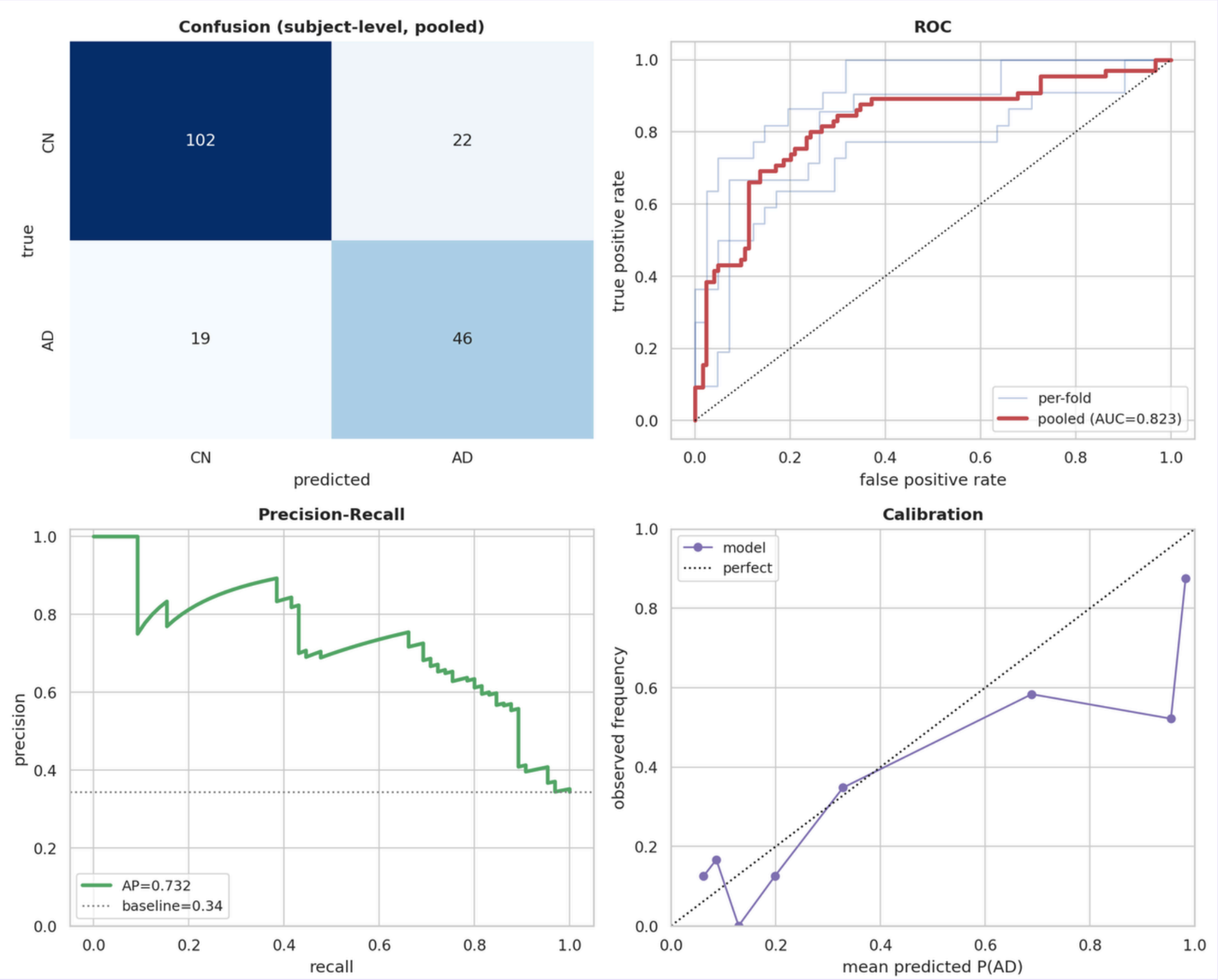
Score-CAM Heatmap for DenseNet121 Binary CNN

XAI COMPARISON FOR BINARY CNN 36



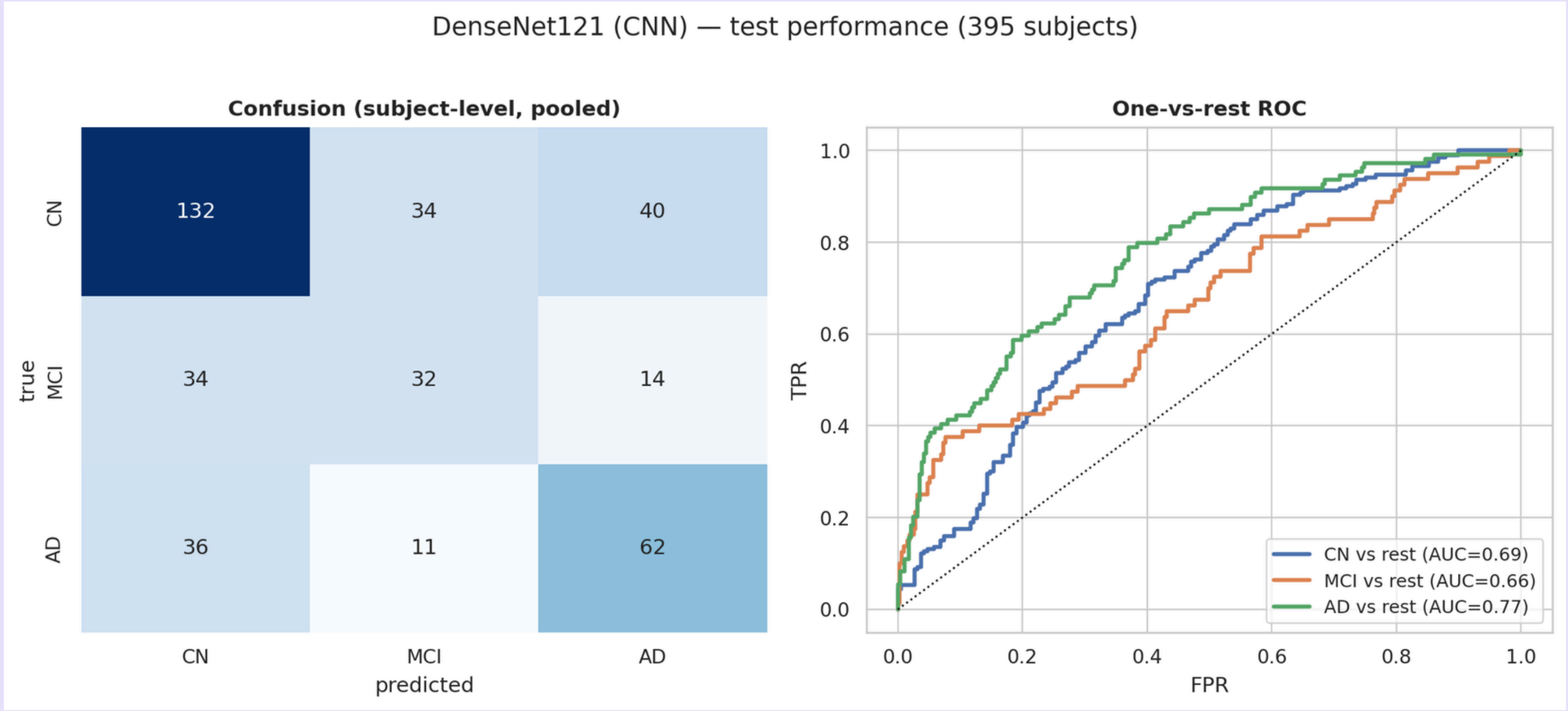
LIME Region-Based Explanation for DenseNet121 Binary CNN

PERFORMANCE DIAGNOSTICS



LIME Region-Based Explanation for DenseNet121 Binary CNN

PERFORMANCE DIAGNOSTICS



3-CLASS — MCI IS THE LOW-AUC CLASS

STAKEHOLDER REQUIREMENTS

Name	Specialization	Years of Experience	Clinical Reliability Opinion	Explainable AI (Grad-CAM) Clinically Relevant?	Helpful for Diagnosis?
Dr. Ziaur Rahman	Physician & Surgeon	12	Reliable	Yes, highly relevant	Yes
Dr.Abul Latif	Radiologist	9	Reliable	Yes, improves interpretability	Yes
Dr. Tania Kabir	Radiologist	16	Reliable	Partially relevant	Yes
Dr. Subir Chandra Das	Neurologist	8	Technically reliable	Yes, essential for model transparency	Yes

EXTERNAL TESTING

Model	Trained for	ADNI 5-fold	Hosp 3-class acc	Hosp CN-vs-MCI acc	Hosp AUC*
Swin	3-class	57.7%	38% (3/8)	75% (6/8)	0.62
CNN (DenseNet)	3-class	57.2%	25% (2/8)	38% (3/8)	0.50
ViT	3-class	28.6%	25% (2/8)	50% (4/8)	0.12
CNN (DenseNet)	binary	81.0%	n/a	75% (6/8)	0.94
ResNet-18 (MedicalNet)	binary	69.8%	n/a	50% (4/8)	0.94

Local Hospital External Testing Results

Hospital data was used as a pilot external test; labels were derived from radiology reports under the guidance of Dr. Ziaur Rahman from Comfort Diagnostic Centre.

CONTRIBUTIONS

- Leakage-free subject-level benchmark using 5-fold cross-validation
- Compared five 3D architectures with identical preprocessing and evaluation
- Reported honest performance using Accuracy and Macro-F1
- Validated predictions with Explainable AI (Grad-CAM, IG, SmoothGrad, Attention)

LIMITATIONS & FUTURE WORK

Limitations

- Small MCI sample size (80 subjects)
- Single-dataset evaluation (ADNI)
- XAI provides coarse localization only

FUTURE WORK

- Multi-center external validation
- Larger and balanced datasets
- Improved hybrid architectures
- Fine-grained 3D explainability

Ps Attribute	How Ps are Addressed in the Project	CLOs	PLOs
P1: Depth of Knowledge Required	Requires advanced knowledge of ML, 3D CNN/ViT architectures, mathematical modeling, and 3D MRI preprocessing. [K3, K4, K6, K8]	CLO1,CLO2 ,CLO7	PLO1, PLO2, PLO5
P3: Depth of Analysis Required	The problem has no obvious solution and requires custom architecture design and fusion strategies [K3]	CLO2, CLO7	PLO2, PLO3, PLO4
P4: Familiarity of Issues	Involves unfamiliar 3D medical imaging tasks, like MRI preprocessing, Alzheimer's classification, subject-level evaluation, and XAI interpretation. [K4, K6, K8]	CLO1, CLO7	PLO4, PLO5, PLO12
P6: Stakeholder Involvement	Various stakeholders involved: healthcare professionals and patients to evaluate model [K5, K8]	CLO2, CLO5, CLO6	PLO6, PLO8, PLO10
P7: Interdependence	Multiple dependent modules-preprocessing, 3D feature extraction, fusion, classification, evaluation, and XAI working together as one complete system pipeline. [K5, K6]	CLO3, CLO7	PLO3, PLO5

As Attribute	How As are Addressed in the Project	CLOs	PLOs
A1: Range of Resources	Used ADNI MRI dataset, GPU hardware, PyTorch, and medical imaging tools such as Nibabel/MONAI.	CLO2, CLO7	PLO3, PLO5
A2: Level of Interaction	Integrated 3D MRI preprocessing, model training, feature extraction, evaluation, and XAI into one workflow.	CLO2, CLO7	PLO2, PLO3, PLO5
A4: Consequences for Society and Environment	Supports Alzheimer's decision-support research by assisting clinicians and improving model transparency.	CLO4, CLO5	PLO6, PLO8
A5: Familiarity	Extends beyond prior experience through 3D MRI processing, Transformer-based modeling, subject-level evaluation, and XAI.	CLO1, CLO7	PLO1, PLO4, PLO12

CONCLUSION

- Hybrid 3D CNN–ViT achieved the best binary performance (82.2%)
- Established a leakage-free subject-level 5-fold evaluation framework
- Explainable AI improved model transparency and interpretability
- Fair comparison of CNN, Transformer, and Hybrid models

References

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THANK YOU